

An explainable artificial intelligence feature selection framework for transparent, trustworthy, and cost-efficient energy forecasting

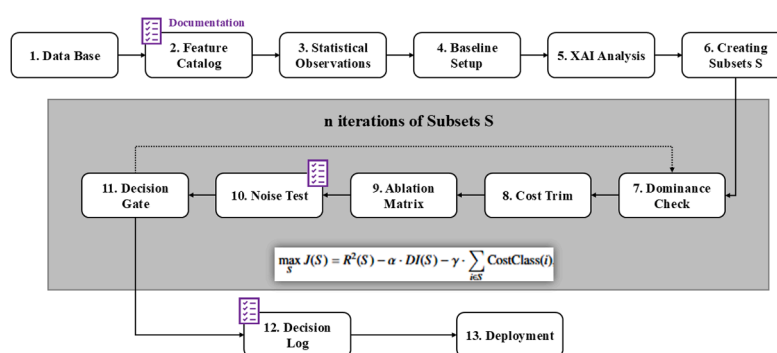
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HIGHLIGHTS

- XAI-driven framework for transparent feature selection in renewable energy forecasting.
- Removing dominant features improves robustness under noise.
- Cost-sensitive feature reduction with minimal accuracy loss.
- Scalable, human-in-the-loop framework for industrial renewable energy forecasting.

GRAPHICAL ABSTRACT



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ABSTRACT

Accurate forecasting of renewable power generation is crucial for grid stability and cost efficiency. Feature selection in AI-based forecasting remains challenging due to high data acquisition cost, lack of transparency, and limited user control. We introduce a transparent and cost-sensitive feature selection framework for renewable power forecasting that leverages Explainable Artificial Intelligence (XAI). We integrate SHapley Additive exPlanations (SHAP) and Explain Like I'm 5 (ELI5) to identify dominant and redundant features. This approach enables systematic dataset reduction without compromising model performance. Our case study, based on Photovoltaic (PV) generation data, evaluates the approach across four experimental setups. Experimental results indicate that our XAI-based feature selection reduces the dominance index from 0.37 to 0.17, maintains high predictive accuracy ($R^2 = 0.94$, drop < 0.04), and lowers data acquisition costs. Furthermore, eliminating dominant features improves robustness to noise and reduces performance variance by a factor of three compared to the baseline scenario. The developed framework enhances interpretability, supports human-in-the-loop decision-making, and introduces a cost-sensitive objective function for feature selection. By combining transparency, robustness, and efficiency, we contribute to the development and implementation of Trustworthy AI (TAI) applications in energy forecasting, providing a scalable solution for industrial deployment.

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1. Introduction

Global energy demand is accelerating due to digitalization, industrialization, and rising living standards. This trend poses a significant challenge: ensuring sustainable, reliable, and affordable energy while minimizing environmental harm. Energy consumption intensifies across all sectors, e.g., from residential and commercial use, transportation, and manufacturing. Rising demand creates the dual challenge to maintain reliability while addressing the environmental impacts of conventional, fossil-based systems [1,2]. Their limits under growing demand underscore the urgency of a clean-energy transition [3]. Energy pathways diverge: many emerging economies still rely on fossil fuels, while other countries increasingly rely on renewable energy [1,4–6].

Variable renewables challenge grid stability [7]. Balancing volatile supply and fluctuating demand requires advances in storage, flexibility, and smart-grid control. Accurate generation and load forecasting are pivotal [8–11]. While statistical models have traditionally been used for forecasting, Artificial Intelligence (AI) is a key technology in the energy sector [12–14]. Due to its capacity to model complex, nonlinear relationships, AI demonstrates high efficacy across a wide range of energy management applications [15]. AI enhances the efficiency of energy generation, distribution, storage, consumption, and trading processes [11,16]. Machine Learning (ML), as a subset of AI, improves the accuracy and efficiency of energy generation forecasting by identifying complex patterns in large historical datasets, often outperforming traditional statistical models [11,16–20]. This is particularly important for integrating volatile renewable energy sources, such as wind power and PV, whose generation and load demand must be accurately forecast to optimize the grid and reduce cost [6,11,15,16,21]. AI streamlines Master Data Management (MDM) processing large datasets and reducing manual feature handling, enabling informed, real-time decisions and cost-effective, sustainable operations [11,17,21–23].

The implementation of AI in energy systems requires careful consideration of the potential risks and challenges associated with its use [20,24,25]. Integrating AI systems and models into energy systems requires high-quality, contextually relevant data, yet ensuring sufficient accuracy, accessibility, and consistency remains a significant challenge [18,26–28]. Improperly structured, erroneous, or biased data reduce accuracy, increase cost, and thus represent a significant challenge for the reliable integration of AI in energy systems [29,30]. Applying feature selection and dimension reduction is crucial for enhancing the performance of AI models and systems and increasing their cost-efficiency [31–33]. XAI has emerged as a key solution for generating interpretable outputs. This improves transparency and enables trust, which is essential for integrating and accepting AI in the energy sector [33–35]. The complexity of models and data involved limits human comprehensibility, which can undermine trust in AI systems [36,37]. TAI is of significant importance in the energy sector, as critical infrastructure places high demands on reliability, security, traceability, and ethical standards [16,20,21,38]. XAI and TAI improve forecasting accuracy and operational efficiency, ensuring compliance with ethical and legal standards. Furthermore, they foster stakeholder acceptance. As a result, XAI and TAI enable the sustainable and effective integration of such systems into critical infrastructures [20,21,38–40,83].

Renewable energy forecasting relies on technical, weather, and historical data [41,42]. Broader datasets increase complexity, risk of overfitting, and cost, making the trade-off between data volume and quality critical. Inaccurate or excessive data collection, as well as the need for continuous provision after implementation, are major cost drivers [30,43,44]. Large datasets further increase workload, operator cost, and computational requirements for acquisition, storage, and personnel [45,46]. In contrast, smaller, high-quality datasets often achieve similar accuracy at lower cost [47]. Feature selection identifies the smallest subset of features with high informational value [48]. Numerous approaches have been developed to reduce dataset size [49,50], yet challenges remain in applying them effectively for cost-efficient

energy forecasting:

- Feature selection methods lack transparency, hindering user understanding and trust; enabling user control can improve acceptance [19,51].
- A single feature can exert disproportionate influence on model forecasting. Standard feature selection methods overlook the potential drawbacks of such dominance, which may compromise model robustness and increase sensitivity to variations in that variable.
- Conventional forecasting methods do not enable users to integrate their expertise into the feature selection process, thereby limiting the potential for hybrid AI [33,52].

XAI enhances transparency by revealing decision-making processes and identifying independent features that are essential to model performance. It enables efficient dataset review, strengthens user control, and promotes model interaction and acceptance. To address the challenges of feature selection, we investigate the following Research Question (RQ):

RQ1: How can XAI methods enable the reduction of features in datasets for energy generation forecasting?

We use XAI to reduce feature sets, lower data complexity, and improve model interpretability. Guided by XAI evaluations, only sub-datasets are used for training, which decreases model complexity, training time, and cost. Eliminating dominant, cost-intensive, and redundant features increases processing efficiency. Dominant features are those that significantly and almost exclusively determine the model's output. In contrast, cost-intensive or redundant features exhibit negligible measurable influence on the model's output and therefore lead to unnecessary cost, rendering them negligible. We apply XAI methods to enhance the interpretability of feature contributions to AI-driven decisions, enabling targeted feature reduction and cost-efficient data adjustments. Given the substantial expenses associated with data curation, limiting the continuous provision of data supports efficiency [53]. XAI methods, such as SHAP and ELI5, provide transparency and explainability to users, enabling them to understand the underlying factors influencing model decisions [54–56]. To ensure the necessary reliability and simplify MDM, the XAI analysis focused on SHAP and ELI5. We selected SHAP based on game theory and its ability to provide consistent local and global explanations of feature contributions [57–61]. ELI5 provides a practical and direct visualization of feature importance, facilitating debugging and enabling insights [62,63]. This ensures efficient, transparent, and trustworthy energy forecasting. Our framework enhances transparency in feature selection and alleviates the cost burden associated with large datasets in industrial contexts.

Feature reduction can slightly decrease model quality, as each excluded feature may still contain valuable information, yet it remains necessary to limit cost and reduce the dominance of single variables [64]. The decrease in model quality is minimal compared to the benefits of the provided information, notably when incorporating other features. Building on the initial exploration of feature reduction through XAI (RQ1), RQ2 and RQ3 examine the implications of this reduction on model performance and the distribution of feature importance.

RQ2: How does removing dominant or cost-intensive features affect model performance and feature relevance?

Despite the observed decrease in model quality, eliminating these features can promote a more robust and equitable distribution of importance among the remaining features. Thus, remaining features will have a more balanced importance in a model's forecasting, resulting in more stable and reliable models that are less susceptible to bias associated with any single feature. Conventional feature selection models without human-in-the-loop interaction cannot achieve this.

RQ3: How can XAI methods support observation and analysis of feature importance distributions to enable an equitable feature allocation?

Inspired by Design Science Research (DSR) based on Hevner [65]

and Schoormann [66], we deduce the transparent and cost-efficient XAI Feature Selection (XAI-FS) Framework using XAI methods that enhance user involvement and model robustness in energy forecasting. Our framework is suitable for implementation as it contains numerous functions for forecasting energy generation, including many features that do not significantly impact the outcome [67] and result in increased cost and power demand for model operators. In our framework, the application of XAI facilitates an identification of features with minimal impact on the model, as well as those with significant influence that can disproportionately affect the model's output. Consequently, in cases where a model exhibits a strong dependency on a single variable, our model has limited robustness. Our framework constitutes a hybrid AI paradigm that actively involves users in critical decision-making processes, ensuring both user control and the effective integration of domain-specific expertise. We explore a model's behavior and remove weak and particularly dominant features. Analyzing scenarios demonstrates how feature selection influences the robustness and reliability of model forecasting while reducing the dataset and number of features. We evaluate our results and findings using additional data and models commonly used in energy forecasting. To address the challenges of interpretability, cost-efficiency, and robustness in PV forecasting. We provide the following key contributions:

We contribute to the theory with an XAI-FS Framework that enhances transparency, reduces data cost, and enables human-in-the-loop decision-making in energy forecasting models. Unlike prior work, our framework systematically integrates XAI-based feature selection with cost-awareness and robustness testing. It operationalizes the removal of dominant features and empirically demonstrates improved robustness under noise. Based on related work in XAI and feature selection in the energy sector, we present our framework, results, and findings. We discuss our practical and theoretical contributions and provide a further research agenda. The remainder of our work is organized as follows. [Section 2](#) provides a comprehensive review of the relevant literature and theoretical foundations concerning XAI and feature selection in the context of energy forecasting. [Section 3](#) outlines the research design, describes the datasets, and specifies the experimental procedures. [Section 4](#) presents the empirical results and insights derived from the proposed framework for feature selection based on XAI. [Section 5](#) critically discusses the implications of the findings, acknowledges methodological limitations, and proposes directions for further research. [Section 6](#) concludes the manuscript by summarizing the principal contributions and identifying potential avenues for further investigation.

2. Literature search and review

2.1. Status quo

We conduct a systematic literature review inspired by vom Brocke et al. [68] and an AI- and graphic-based literature search. We searched the databases SpringerLink, Web of Science, Elsevier, AIS Electronic Library, IEEE Explore, ACM, and Wiley with the search terms 'AI' OR 'Artificial Intelligence' OR 'ML' OR 'Machine Learning' AND 'Explain*' OR 'Transparency' OR 'Respons*' OR 'Trust*' OR 'Human' OR 'HCI' OR

'HMI' AND 'Feature' OR 'Feature Selection' OR 'Feature Reduction' OR 'Data' AND 'Energy' OR 'Renewable Energy' in the period from January 2020 to July 2025. After reviewing titles, abstracts, and conclusions, we selected 54 publications. After full-text review, 12 publications were excluded. Based on Webster and Watson [69], we conducted a forward, backward, and author search. Five publications were added. Ten publications were added via a Google Scholar similarity search and an AI- and graphic-based search in Semantic Scholar, SciSpace, and ORKG Ask databases. We identified exclusion criteria to ensure high-quality publications. The exclusion criteria include results and findings, impact factor, number of citations, and the added value of each publication. We included 57 publications in our final literature dataset. Selection was based on added value, quality, impact, Scimago factors, and citation metrics.

Accurate energy consumption forecasting enables proactive resource allocation, cost reduction, and operational efficiency [22,70]. The complexity and uncertainty of energy system operations have increased significantly with access to intermittent and decentralized generation sources and the development of electricity markets. The continuous generation of high-dimensional, heterogeneous data sources poses a new challenge for smart grid management [71]. Feature selection distinguishes relevant from less important variables, enhancing the accuracy, efficiency, and interpretability of ML models. In high-dimensional datasets, feature selection reduces computational load, especially during iterative development [19,22,52,60]. A clear representation of feature–output relationships enables users to assess the validity of forecasting and fosters justified trust [72]. Feature selection methods are usually divided into three categories: filter methods, wrapper methods, and embedded methods. Filter methods are efficient and rely on data properties. Wrapper methods are performance-based but computationally intensive. Embedded methods are integrated into the training process and rank features during learning [19,22].

XAI methods do not fit strictly into traditional categories; instead, they employ an interpretability-based approach that extends beyond forecasting accuracy, offering computational advantages and enabling analysis of decision-making when features are removed [19,73]. Users can optimize resource allocation while maintaining model performance, understanding how removing certain features of cost-intensive yet crucial data points impacts a model's forecasting [74]. Clarifying influential inputs and supporting error detection, XAI fosters user confidence and trust, defined as the willingness to rely on AI despite uncertainty [52,55,63,75]. Trustworthiness refers to the system's consistent and reliable performance, as demonstrated in energy forecasting [34,35]. However, trustworthiness is also subjective and depends on the user's perception and assessment [76]. TAI is crucial to ensure AI models and systems used in energy supply forecasting are powerful, as well as responsible, transparent, and robust [11,20,38]. TAI includes both socio-technical aspects, such as transparency and non-discrimination [77–79,83], and cognitive dimensions, such as explainability and robustness [80,83].

In energy supply forecasting, critical infrastructures require high reliability, security, and ethical responsibility. XAI helps assess the relevance of inputs and makes the factors influencing accuracy transparent, thereby reducing costly errors [39,58]. TAI complements this,

Contributions

1. We introduced a transparent, user-in-the-loop feature selection framework for PV forecasting that integrates SHAP and ELI5 to enhance interpretability and trust in AI-based energy forecasting.
2. Our framework introduces a dominance index and a cost-aware feature reduction strategy, enabling significant savings in data acquisition cost while maintaining predictive accuracy ($\Delta R^2 \leq 0.037$).
3. Through DSR and robustness tests across multiple datasets, we demonstrate that removing dominant features improves model stability under noise without compromising performance.

ensuring a responsible, robust, and trustworthy application, which is vital as forecasting directly affects operations, financial risks, and the reliability of generation [11,20,21,38,39,81,83]. Since traditional ML models are often black boxes that limit trust and practical use in energy systems, XAI provides methods to make models transparent and understandable [22,36,38,39,59,63,71,82,83]. For instance, the DARPA XAI program develops glass box models that are explainable to a human-in-the-loop while maintaining high performance [38,63,84–86]. Dimitriadis et al. [6] use a DL-based approach to PV power forecast, leveraging data from several countries to generalize and improve prediction accuracy. The distribution shift is solved through a three-stage training process and the use of different architectures. The distribution shift arises from the use of heterogeneous data and describes the significant change in data characteristics caused by training DL models for PV power prediction. The ability to explain decisions is essential for building trust and ensuring acceptance of AI in sensitive domains such as the energy sector [38,39,72,87].

Our literature review revealed that only a limited number of publications address feature selection in the context of XAI. Sim et al. [83] developed methods for input variable selection in XAI for energy consumption forecasting and applied them to Korean datasets, thereby improving accuracy. Yaprakdal and Arisoy [22] developed a hybrid medium-term electricity consumption forecasting model that combines feature selection and extraction with three Long Short-Term Memory (LSTM) architectures, and enhanced interpretability by applying the XAI method LIME, which had not been previously used in this area. Maarif et al. [59] developed an LSTM-based energy consumption forecasting model and utilized XAI for parameter analysis. Applying it to a steel company dataset, they achieved low error values and identified key factors to support efficient resource planning. Van Zyl et al. [19] examined gradient-weighted class activation mapping and SHAP for feature selection in national load forecasting using a multi-headed conventional neural network, demonstrating improved forecasting accuracy while also highlighting limitations that require further research. None of these publications investigates feature selection and reduction in the field of XAI using Extreme Gradient Boosting (XGBoost).

XGBoost outperforms models such as Linear Regression (LR) and neural networks [58,70]. Its scalability makes it suitable for large energy datasets, an important aspect for simplifying MDM [58,60,70]. XGBoost is a powerful gradient boosting model used in numerous energy forecasting applications that can outperform other AI models, such as LR and neural networks, in terms of forecasting accuracy [19,58,63,81,83]. MDM refers to the quality, preparation, and governance of data to ensure consistency, accuracy, completeness, and reliability [28,70]. AI performance and reliability depend on the quality of the dataset [28]. Trustworthy forecasting requires input data with a distribution similar to that of the training data [39]. MDM can ensure the quality dimensions of data, which include accuracy, accessibility, consistency, relevance, timeliness, traceability, and usability. This includes identifying and handling outliers [28,59,63,83]. Feature selection as preprocessing removes irrelevant features, creating simpler models, and improving performance through cleaner data [17,19,22,49,88]. This reduces dimensionality and helps counteract problems such as the ‘curse of dimensionality’ and overfitting, which can impair model performance [49,89]. Without effective data management, challenges such as inconsistencies, missing values [23], heterogeneous formats, bias, or concept drift [70] can reduce the accuracy and reliability of forecasting [14,28,39] and necessitate advanced detection and analysis approaches [28]. Although XGBoost offers basic interpretability via decision trees, combining it with XAI methods deepens explanations [36,58].

The integration of SHAP in XGBoost enhances the model’s interpretability by analyzing the interactions and dependencies among predictors and the target variable [58]. To improve the explainability of XGBoost models and support feature reduction, various XAI methods are often applied, e.g., SHAP, LIME, and ELI5 [22,38,39,63].

2.2. Theoretical framework

We will examine the two XAI methods SHAP [54,90] and ELI5 [56,62]. SHAP is based on concepts from cooperative game theory and quantifies the contribution of each feature to a model’s output. It provides both local and global explanations [38,39,59,60,63,81]. SHAP values support feature selection by identifying important variables and improving model accuracy [19,38,59,63,83]. ELI5 is a Python library that provides global feature importance for ML models such as XGBoost. Removing features based on ELI5 reduces dataset size and improves accuracy, though it may risk performance loss [38,63]. LIME, a model-agnostic XAI method, explains individual forecasts using local surrogate models. While applicable to XGBoost, LIME provides only local explanations [22,38,63,71].

Global explanations summarize the impact of features across all instances, revealing which features most influence overall forecasting [91]. In contrast, local explanations examine individual cases, showing how specific features affect a single prediction [92].

SHAP measures the impact of individual features by calculating their average marginal contributions [57]. ELI5 improves transparency by providing intuitive explanations for complex ML models [56]. In contrast, LIME is less reliable because it ignores feature correlations and lacks the theoretical guarantees of SHAP [36,61]. Overall, SHAP offers deep, theoretically grounded insights, while ELI5 provides fast and easy-to-understand feature importance analysis [36,61,62].

2.2.1. Shapley additive explanations

SHAP quantifies each feature’s contribution to individual forecasts. It provides consistent and interpretable explanations of complex ML decision processes. SHAP applies Shapley values to ensure local accuracy, meaning that feature attributions sum to the model output. It also ensures consistency, as a feature’s contribution increases with its influence. These properties allow for meaningful comparisons of feature importance [57,73,89,93]. SHAP uses Shapley values to assess each feature’s contribution to overall forecasting. It calculates the average change in forecasts when a feature is added to different feature combinations. These values indicate both the global significance of features and their impact on individual forecasts [57]. TreeSHAP is specifically optimized for tree-based models such as XGBoost. It enables efficient computation of Shapley values and captures feature interactions with higher accuracy. In the energy sector, SHAP is frequently used to explain ML forecasts. It also serves to identify and compare the importance of individual features across different forecast instances [81]. Our approach ensures a fair and consistent attribution of feature importance, thereby enhancing the interpretability and reliability of model forecasting:

$$\phi_i(f, x) = \sum_{S \subseteq X \setminus \{i\}} \frac{|S|!(|X| - |S| - 1)!}{|X|!} [f(x_S \cup \{i\}) - f(x_S)] \quad (1)$$

Here, X is the set of input features, and S is any subset of X excluding feature i . The term $f(x_S \cup \{i\})$ denotes the model’s forecasting when feature j is included, while $f(x_S)$ represents the forecast when only features in S are considered. Computing exact Shapley values requires extensive computation, up to 2^I evaluations.

Exact Shapley values require extensive computation, so they are typically approximated using the dataset and model. These approximations, while not perfect, are sufficiently accurate for practical use. We apply TreeSHAP, which is optimized for tree ensembles and delivered the best results in our model selection. Alternative methods such as KernelSHAP and GradientSHAP are therefore excluded. TreeSHAP reduces complexity from 2^I to $O(TLD^2)$, where T represents the number of trees, L the maximum number of leaves, and D the tree depth, which scales logarithmically with L . To optimize computation for tree-based models, Eq. (1) is reformulated to match the decision tree structure. TreeSHAP tracks each feature’s contribution by following paths through

the tree and calculating the probability that an observation passes each node. The adapted equation is:

$$\phi_i(f, x) = \sum_{i=1}^M \left(\sum_{S \subseteq X \setminus \{i\}} \frac{|S|!(|X| - |S| - 1)!}{|X|!} [f(x_S \cup \{i\}) - f(x_S)] \cdot P_i(x) \right) \quad (2)$$

Here, M is the number of trees in the ensemble, and $P_i(x)$ is the probability that instance x follows a specific path in tree i , based on the subset S . This structure avoids computing all feature permutations by leveraging tree paths and probabilities [57,94].

2.2.2. Explain like I'm 5

ELI5 provides global explanations of feature importance, clarifying factors that influence a model's decision-making across the entire dataset. It outputs direct feature weights and visualizes forecasting as the sum of contributions plus bias. ELI5 is compatible with libraries such as XGBoost [62]. The method employs Permutation Feature Importance (PFI), which quantifies feature relevance by measuring the performance drop when feature values are randomly shuffled, thereby revealing their contribution to forecasting [39]. Consider a forecasting model m applied to a dataset X . The performance of m on X is evaluated using a specified metric, such as Root Mean Square Error (RMSE), denoted as S . RMSE is a widely used metric for evaluating model performance that quantifies the average size of forecasting errors [7,22,59]. It is calculated as the square root of the mean of the squared errors and penalizes larger deviations disproportionately. A lower RMSE value is desirable and indicates better accuracy, which is why it is used to evaluate and optimize AI models in areas such as energy consumption forecasting, PV energy generation, and electricity price forecasting [7,14,15,22,23,42,59,83,95,96]. R^2 indicates the proportion of the total variance in the target variables explained by the model [7,61]. R^2 is a measure of the overall model fit and reproducibility [97]. This result serves as the baseline for assessing the importance of permutation. Now, for each feature i , the respective values are randomly changed for all entries k , and thus, a corrupted version of the dataset x is generated, called $X_{k,i}$. The quality $S_{k,i}$ is calculated again for this falsified dataset. Thus, the PFI j_i for the feature i is calculated as [56]:

$$j_i = S - \frac{1}{X} \sum_{k,i=1}^X S_{k,i} \quad (3)$$

2.2.3. Further metrics for comparison

In addition to PFI, Shapley values, normalized results, and statistical measures such as standard deviation, two further metrics are used to evaluate models: the coefficient of determination (R^2) and RMSE [98,99]:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (4)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (5)$$

n represents the total number of forecasts made by a model. For each observation j , where i ranges from 1 to n , $\hat{y}_i$ denotes the forecast value generated by the model, while y_i signifies the actual observed value. Additionally, $\bar{y}$ represents the mean of all observed values across a dataset. This notation systematically compares forecast outcomes with actual observations and evaluates the model's performance [100].

Using these metrics on the original dataset, we select the best-performing model from the tested candidates. This model is then used exclusively in subsequent analyses. In the main phase, the same metrics are employed within the context of a feature ablation test. This test systematically removes features in descending order of importance, as determined by XAI methods, and reassesses the model's quality after each iteration. The model's performance is expected to degrade

progressively as more features are removed. Removing individual features has a minor impact on the model's performance, as other features can compensate for these losses. If more features are removed, especially those considered important by XAI, a more significant decrease in model quality can be expected. This reduction underscores the substantial contribution of these features to the model's predictive accuracy, as evident in the evaluation metrics, including the R^2 score and RMSE. Thus, the feature ablation study provides a robust framework for understanding the relative importance of different features and their contribution to the model's overall performance. By systematically evaluating the impact of feature removal, researchers and practitioners gain deeper insights into model behavior and the data patterns it captures. This process enhances the interpretability and reliability of forecasting.

3. Research design

3.1. Design science research

We conduct a DSR-inspired approach based on Hevner [65] and Schoormann [66]. The research design, illustrated in Fig. 1, begins with an Entry Phase. In Step 1, the problem is defined: traditional feature selection lacks transparency, fails to account for dominant features, and leads to inefficient data utilization. Step 2 introduces a conceptual framework that guides the use of XAI methods. It integrates human-in-the-loop approaches, improves data efficiency, and identifies both dominant and redundant features. The Experimental Phase commences with Step 3, which examines datasets for statistical characteristics. Dataset 1 receives primary attention, while the other datasets are used for validation. In Step 4, a suitable model is selected for conducting energy forecasting based on Dataset 1. Step 5 involves analyzing features using SHAP and ELI5, including a comparative evaluation against traditional feature selection methods. In Step 6, a structured test design is implemented.

Setup 1 defines the baseline in which the dataset remains unchanged. Setup 2 removes the dominant feature from the dataset. Setup 3 excludes all redundant features. Setup 4 eliminates both the dominant and all redundant features. Additionally, an initial robustness test is conducted between Setup 1 and Setup 2 to demonstrate that the robustness of Setup 2 significantly improves in the absence of the dominant feature. In Step 7, this robustness test is expanded and refined, with a detailed evaluation of metrics and residuals. The Closure Phase includes Step 8. Results are discussed, and a framework is developed for both academic and practical applications. Finally, in Step 9, all results and findings are consolidated for dissemination to the scientific community and practitioners. Fig. 1 depicts our research process graphically. Steps 1–8 are explained with their activities, and Steps 3–6 contain our results and findings for the associated steps.

3.2. Dataset and model

The datasets used in this study are provided in the online appendix and were sourced from the Kaggle platform.¹ Dataset 1, which serves as the reference dataset, pertains to a single PV system consisting of multiple modules [101]. In contrast, both validation datasets focus on forecasting renewable wind energy. As these datasets are used exclusively for validation purposes, the focus of this section is placed on Dataset 1.

Dataset 1 comprises energy generation data recorded at 15-minute intervals, along with various meteorological parameters specific to the PV system's location in Poland. Although no particular region was targeted during data collection, the Polish dataset proved to be the most

¹ <https://www.kaggle.com/datasets/samanemami/renewable-energy-and-weather-conditions>

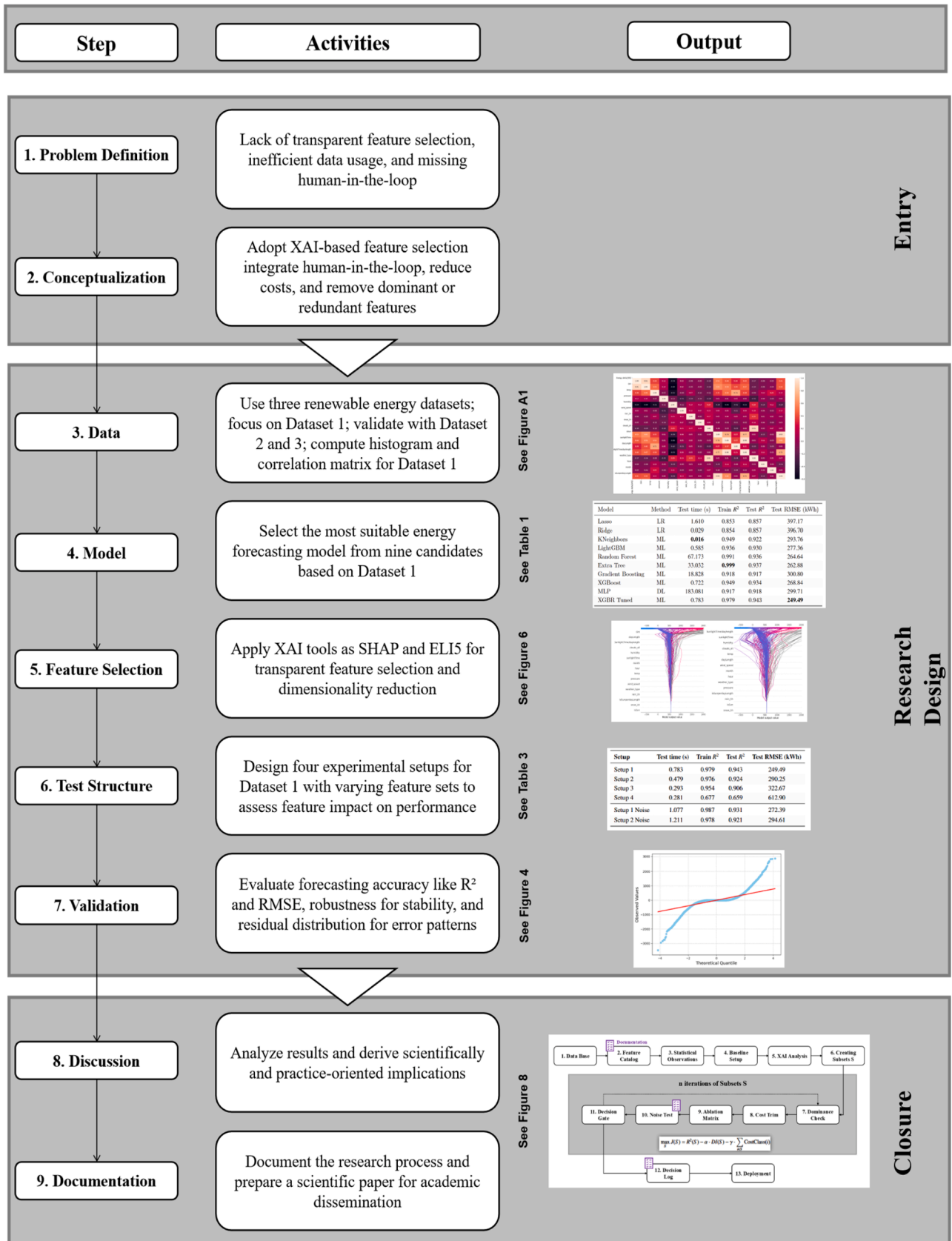


Fig. 1. Research process based on DSR [66].

suitable among the available alternatives. The dataset spans a total of 2069 days, yielding 198,624 individual data points. It includes timestamps, energy output values, and 15 distinct features, such as GHI, which quantifies the total solar irradiance incident on a horizontal surface. The extensive temporal coverage, combined with the large number of data points and the diversity of features, makes this dataset suitable for performing feature selection analysis. Due to its exceptional depth and quality, all methodological approaches are applied within the scope of four case studies. To gain a preliminary understanding of the dataset, we analyzed the distributions of each feature and its correlations, including their relationships with the target variable. The analysis of the features Energy Delta [Wh], GHI, Sunny, SunlightTime, and SunlightTime/DayLength indicates a strong correlation due to their similar distributions, as shown in Fig. 2. This observation supports a logical relationship between sun exposure, irradiation, and energy generation in PV systems. In analyzing the correlation between features, certain features exhibit a high correlation, as shown in Fig. A.9.

For instance, DayLength and temperature indicate a strong correlation, as do features derived from combinations of other features, which are inherently related. This implies that some features contain redundant or nearly identical information, necessitating their reduction [48, 102]. Our feature reduction is deferred to a subsequent phase, as feature selection is performed using XAI methods to ensure increased transparency and interpretability.

An appropriate model must be selected to process the dataset effectively. To this end, we evaluated nine models based on recommendations from the scientific literature on suitable regression approaches [103,104]. These models encompass statistical methods such as simple LR [105,106], ML [107–112], and Deep Learning (DL) approaches [6, 113]. The selected models are those most commonly used in regression analysis. DL models were generally excluded due to their increased complexity, computational requirements, and associated cost. Nevertheless, one representative model from this category was included for illustrative purposes.

A comparative analysis shows that Extreme Gradient Boost for Regression (XGBoost) delivers superior predictive performance while maintaining low computational overhead (see Table 1). XGBoost is a sophisticated ML model that utilizes ensembles of decision trees within a gradient boosting framework. In our framework, weak learners are iteratively trained to minimize forecasting errors, thereby improving

Table 1

Performance comparison of models on the target task.

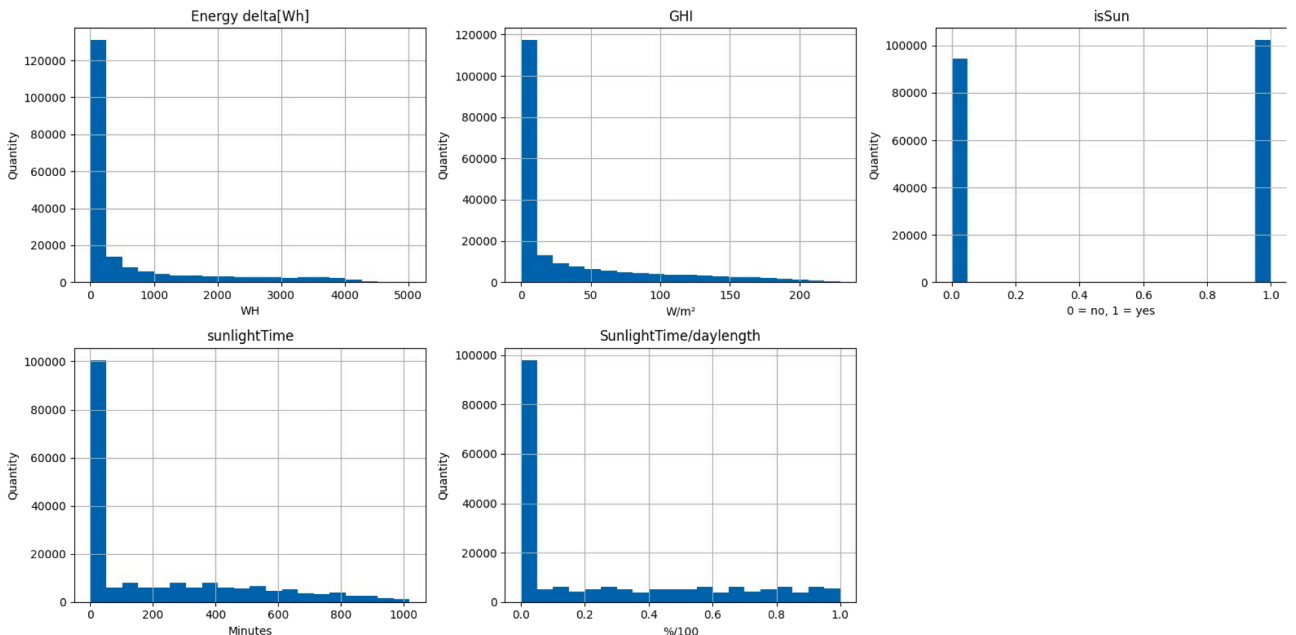
Model	Method	Test time (s)	Train R^2	Test R^2	Test RMSE (kWh)
Lasso	LR	1.610	0.853	0.857	397.17
Ridge	LR	0.029	0.854	0.857	396.70
KNeighbors	ML	0.016	0.949	0.922	293.76
LightGBM	ML	0.585	0.936	0.930	277.36
Random Forest	ML	67.173	0.991	0.936	264.64
Extra Tree	ML	33.032	0.999	0.937	262.88
Gradient Boosting	ML	18.828	0.918	0.917	300.80
XGBoost	ML	0.722	0.949	0.934	268.84
MLP	DL	183.081	0.917	0.918	299.71
XGBR Tuned	ML	0.783	0.979	0.943	249.49

Best values per column in bold. A lower RMSE is preferable; a higher R^2 is more desirable. Test time refers to the total inference time.

overall model accuracy. The model supports extensive parallelization and integrates regularization methods, which enhance its scalability and robustness when applied to large and complex datasets. These capabilities enable preliminary automated feature selection by assigning lower weights to less relevant features. However, these mechanisms are insufficient for comprehensive feature selection, necessitating the use of additional feature selection procedures [112].

Despite higher R^2 values observed in the Random Forest and Extra Tree models, a substantial decline in predictive performance on previously unseen test data compromises their reliability. This degradation indicates a pronounced tendency toward overfitting, suggesting that these models capture noise rather than underlying data structures, thereby limiting their generalizability [114].

In contrast, XGBoost demonstrates notable computational efficiency, operating approximately 60–120 times faster than the aforementioned models, as shown in Table 1. This considerable reduction in processing time is particularly advantageous in contexts requiring rapid model deployment and iterative development. The marginal difference in R^2 scores is acceptable given the much faster processing and the consistent R^2 ratio between training and test data, indicating greater stability and robustness [98]. The Light Gradient Boosting Machine (LightGBM) constitutes a viable alternative, exhibiting negligible differences in predictive quality. The models' performance is comparable to XGBoost,

**Fig. 2.** Histogram of correlated features.

positioning them as close competitors, particularly in applications where minor deviations in predictive accuracy are critical. Nevertheless, the simplicity and computational speed of XGBoost continue to offer practical advantages. In contrast, simple LR models such as Lasso exhibit inferior performance. Their inability to capture complex, nonlinear relationships within the data results in significantly lower predictive metrics, rendering them unsuitable for tasks that demand high accuracy and robustness. After optimizing the hyperparameters of the XGBoost model through 24,192 iterations using RandomizedSearchCV [115], the test R^2 value improved to 0.943.

3.3. Test structure

We examine four test cases to show how XAI methods clarify the influence of individual features on model outcomes and support informed feature selection. In the following, these four cases are referred to as setups. These setups differ in the number of selected features, which were chosen based on their reduction through XAI, either due to their classification as cost-intensive redundant features or as dominant features. All features with a baseline value below 0.020, as determined by XAI-based feature importance analysis, were identified as redundant. The feature GHI was classified as dominant because its importance value in Setup 1 is about 14 times higher than that of the second most influential feature. Our framework simulates how the influence of features on model performance varies across different feature sets. This process enhances the transparency and interpretability of feature selection.

All test setups are detailed in Table 3. In the initial Setup 1, the model is trained using the complete feature set and serves as the baseline configuration. This process is examined using two XAI methods.

The feature importance values derived from these methods, in conjunction with statistical methods such as the correlation matrix, form the basis for selecting the subsequent three setups.

In Setup 2, the most dominant feature is excluded, considering only important and redundant features. Setup 3 removes negligible features, retaining the most dominant and important features. In Setup 4, the most dominant and negligible features are excluded, focusing solely on important features. The structure of our test is listed in Table 2.

To test robustness, noise was systematically added to each setup, targeting the most significant feature identified by feature importance analysis. In this process, for Setup 1 and Setup 2, 20 iterations were conducted in which between 10 % and 40 % of all values of the respective dominant feature were deliberately altered. Care was taken to ensure that the degree of alteration remained identical across both setups to maintain comparability [116].

The procedure for identifying dominant and redundant features using XAI methods was also applied to Datasets 2 and 3 to evaluate the framework. For both datasets, only a reduced model performance was observed, which can be attributed to the use of basic models that were not subject to hyperparameter tuning. Nevertheless, XAI methods enabled the identification of dominant and redundant features that could not be detected using conventional feature selection methods. Similar experimental setups were constructed for both datasets to evaluate the framework.

4. Results and findings

4.1. Models

With an R^2 value of 0.9435 in the validation run, the selected model outperforms all other observed models, demonstrating its suitability for solar plant forecasting. The approximate difference of 0.03 in R^2 between the training and validation datasets indicates that the model can generate meaningful forecasts for previously unseen weather data.

The residual distribution for Setup 1, as illustrated in Fig. 3, shows a reasonable spread with a pronounced concentration of values around zero. This distribution deviates slightly from a normal distribution,

Table 2

Feature importance across four setups.

Feature	Category	Setup 1	Setup 2	Setup 3	Setup 4
GHI	dominant	1.523 ± 0.0074	NVR	2.083 ± 0.015	NVR
Hour	important	0.011 ± 0.0002	0.030 ± 0.0004	0.081 ± 0.001	0.038 ± 0.0009
DayLenght	important	0.108 ± 0.0022	0.238 ± 0.0017	0.098 ± 0.0024	0.208 ± 0.0112
SunlightTime	important	0.020 ± 0.0001	0.163 ± 0.0015	0.966 ± 0.0060	0.717 ± 0.0061
ST per DL	important	0.051 ± 0.0005	0.701 ± 0.0085	0.187 ± 0.0013	0.691 ± 0.0064
Reg.					
Sunny	cost-int.	0.000 ± 0.0000	0.000 ± 0.0000	NVR	NVR
Clouds	cost-int.	0.019 ± 0.0003	0.139 ± 0.0014	NVR	NVR
Humidity	cost-int.	0.016 ± 0.0003	0.600 ± 0.0025	NVR	NVR
Month	cost-int.	0.009 ± 0.0003	0.028 ± 0.0004	NVR	NVR
Pressure	cost-int.	0.011 ± 0.0003	0.039 ± 0.0005	NVR	NVR
Wind Speed	cost-int.	0.012 ± 0.0002	0.042 ± 0.0005	NVR	NVR
Weather Type	cost-int.	0.002 ± 0.0002	0.010 ± 0.0006	NVR	NVR
Rain	cost-int.	0.004 ± 0.0002	0.005 ± 0.0002	NVR	NVR
IS per DL	cost-int.	0.008 ± 0.0001	0.028 ± 0.0004	NVR	NVR
Snow	cost-int.	0.001 ± 0.0000	0.0002 ± 0.0001	NVR	NVR

NVR = Not Visible/Relevant. Feature not present in this subset due to cost reduction, redundancy, or dominant feature elimination.

ST = SunlightTime, IS = IsSun, DL = DayLength, cost-int. = cost-intensive.

suggesting that the model achieves high predictive accuracy with predominantly small error magnitudes. However, the distribution remains irregular, indicating inaccuracies, particularly in the extreme regions of ± 100 , where the number of residuals exceeds expectations under a normal distribution and deviates from an optimal error structure.

Quantile-quantile plots confirm this observation, revealing more pronounced deviations in the tails of the distribution. These findings explain the deviation of the R^2 value from 1, as the presence of extreme residuals negatively impacts the overall model performance.

Despite these limitations, the selected model yields optimal results and effectively forecasts the PV system's energy output using weather data. This model is therefore identified as suitable and adopted as the baseline Setup 1 for further analysis with XAI methods and for identifying dominant and optimal features.

In Setup 2 and Setup 3, R^2 values vary only slightly when dominant or cost-intensive features are removed (see Table 3). Specifically, the setup excluding cost-intensive features yields an R^2 value of 0.906, whereas the setup excluding the most dominant feature results in a slightly higher R^2 value of 0.934. This corresponds to a maximum performance deterioration of approximately 0.037. Notably, excluding cost-intensive features previously identified as having minimal impact results in a slightly greater reduction in predictive accuracy than excluding the dominant feature, which had previously exerted a substantial influence on the model outcome.

In the fourth setup, a substantial decline in model performance is observed, with an R^2 value of 0.66, rendering the model largely ineffective. While partial omission of either the most dominant or the redundant feature results in only marginal changes in predictive accuracy, simultaneous removal of both features leads to a pronounced deterioration in model performance. This outcome indicates that the model relies on the combined presence of both features to capture essential variance in the data. Their joint exclusion results in underfitting, as the model is no longer able to adequately represent the

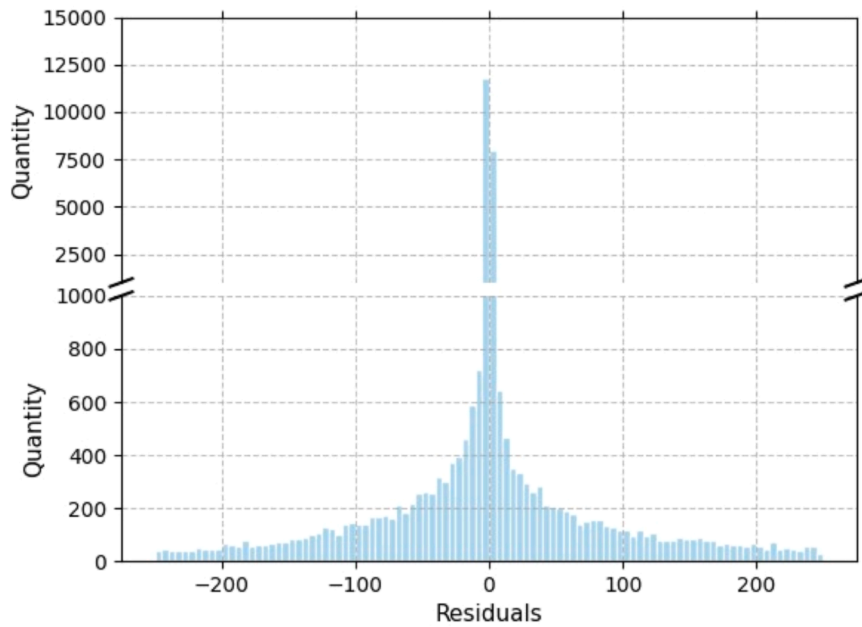


Fig. 3. Histogram of residuals.

Table 3
Results of experimental setups.

Setup	Test time (s)	Train R^2	Test R^2	Test RMSE (kWh)
Setup 1	0.783	0.979	0.943	249.49
Setup 2	0.479	0.976	0.934	290.25
Setup 3	0.293	0.954	0.906	322.67
Setup 4	0.281	0.677	0.659	612.90
Setup 1 Noise	1.077	0.987	0.931	272.39
Setup 2 Noise	1.211	0.978	0.921	294.61

underlying data structure. Fig. 4

When comparing two experimental setups subjected to equivalent noise levels, we observe distinct outcomes. In Setup 1, the coefficient of determination R^2 exhibits a decrease of 0.012 relative to the

uncontaminated configuration. Concurrently, the RMSE increases by approximately 23, resulting in a value of 272.39. In contrast, Setup 2 demonstrates a more modest change in R^2 , with a reduction of 0.003. The RMSE in this setup increases by 4, culminating in a final value of 294.61, as seen in Table 3.

To classify traditional feature selection approaches, filter, embedded, wrapper, and hybrid methods were systematically evaluated using a benchmark dataset. A Gradient Boosting Regressor without feature reduction served as the baseline model. Filter methods, including SelectKBest, ranked features based on univariate statistical scores.

Embedded methods such as LassoCV and Random Forest employed model-inherent importance measures. Wrapper and hybrid methods optimized feature subsets through cross-validation procedures [117, 118]. The empirical results demonstrated that, under conditions of low dimensionality and strong predictive models, classical feature selection

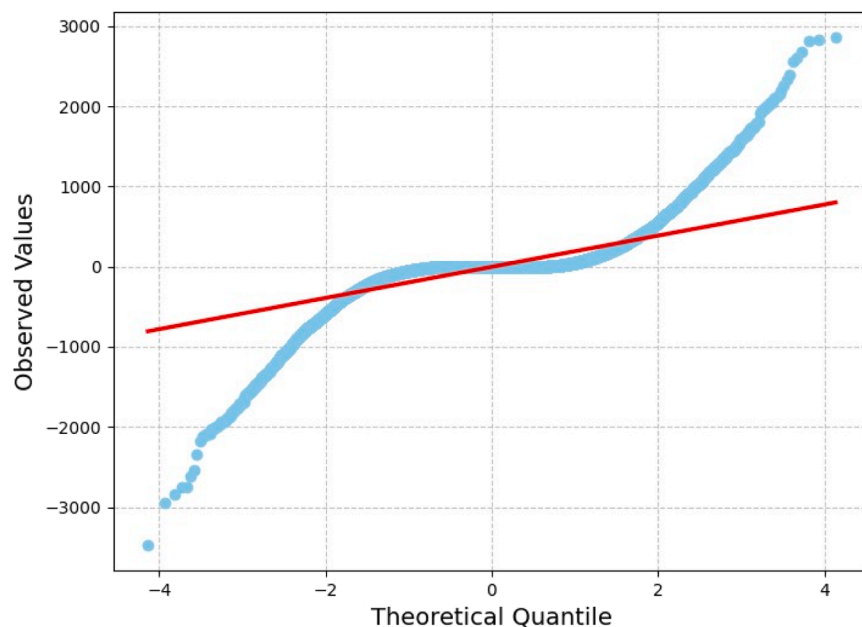


Fig. 4. Quantile-quantile diagram of residuals.

yielded only marginal performance gains. Tree-based embedded methods achieved the highest validation R^2 scores; however, the improvements over the baseline Setup 1 remained limited, with a score of 0.938. Moreover, the analysis revealed no adverse effects associated with the dominance of the feature GHI, indicating that its influence did not compromise model robustness or predictive stability.

For the two datasets employed to evaluate our framework, Random Forest was identified as the most effective baseline model for Dataset 2, whereas XGBR yielded the best performance for Dataset 3. In both cases, the application of XAI methods enabled the consistent identification of dominant and redundant features. Specifically, Dataset 2 includes the feature windspeed_100m, and Dataset 3 the feature WindSpeed, both of which exert a substantial influence on model forecasting. The removal of these dominant features led to only a marginal decline in model performance, indicating their strong but not exclusive predictive relevance. Subsequent XAI-based analyses revealed that, in the respective second experimental setups, feature importance was more evenly distributed across the remaining input variables, further supporting the robustness of the framework.

4.2. Explainable artificial intelligence

The feature importance analysis conducted using the XAI method ELI5 provides a detailed overview of the weightings and their deviations across all features, as presented in Table 3. The initial assessment of the feature weightings indicates that the base dataset is predominantly influenced by a single feature, which substantially determines the forecasting outcome. The cumulative contribution of all remaining features relative to GHI constitutes only 17.8 % of the overall result. Each of the three dataset modifications effectively reduces this pronounced imbalance. Although the weight of the most dominant feature increases in Setup 3, the overall distribution of feature importance becomes more equitable. Setup 2 exhibits a particularly balanced distribution, successfully eliminating the dominance of a single feature observed in Setup 1. Nevertheless, this configuration still contains features with disproportionately high or low contributions to the model outcome. Setup 4 used an aggressive ablation strategy, removing the most predictive variable and most low-contribution features. This reduction substantially compressed the feature space, resulting in clear indications of underfitting, as evidenced by a decline in the coefficient of

determination on the training set to an R^2 of 0.677. The remaining dominant features originate from a single data source and exhibit high intercorrelation, effectively reducing the model to a near one-dimensional signal with limited discriminative power and poor generalizability. Given these minimum-performance and robustness concerns, Setup 4 does not comply with the predefined stopping criteria and is therefore excluded from further analysis. For consistency, the feature Sunny is retained across all setups as a control to account for the physical coupling between irradiance and PV output, but it is not interpreted as an independent causal factor.

Further analysis of feature importance reveals that the ranking of dominant and less dominant features varies whenever subsets of the dataset are generated and different features are excluded. The remaining features either retain their original positions or exhibit slight shifts in rank compared to the full dataset. For instance, the relative importance of the feature SunlightTime increases by >45 , despite the continued presence of the dominant feature GHI in both Setup 1 and Setup 3.

A comparison of the feature importance rankings in Setup 1 and Setup 2, as illustrated in the two diagrams in Fig. 5, reveals substantial differences. Both diagrams display the SHAP representation of identical data points, offering a cumulative local perspective based on 200 model outputs. The left diagram corresponds to the baseline Setup 1, while the right diagram represents Setup 2, in which the most dominant feature has been removed. In Setup 2, most data points strongly influence the output before GHI is added.

In Setup 1, GHI dominates, making its diagram strand much narrower.

Fig. 6 presents the normalized importance values across all four setups. It is evident that Setup 2, which excludes the dominant feature GHI, exhibits a more balanced distribution. Not only do two features exert a significantly stronger influence on the model outcome, but the features SunlightTime, Clouds, and DayLength also gain substantial weight. This is reflected in a lower standard deviation of 0.23, which correlates with greater robustness in Setup 2.

4.3. Robustness test

To highlight the robustness advantage of a balanced feature distribution, the dominant feature in Setup 1 and Setup 2 was deliberately corrupted to the same extent. Specifically, between 20 % and 40 % of the

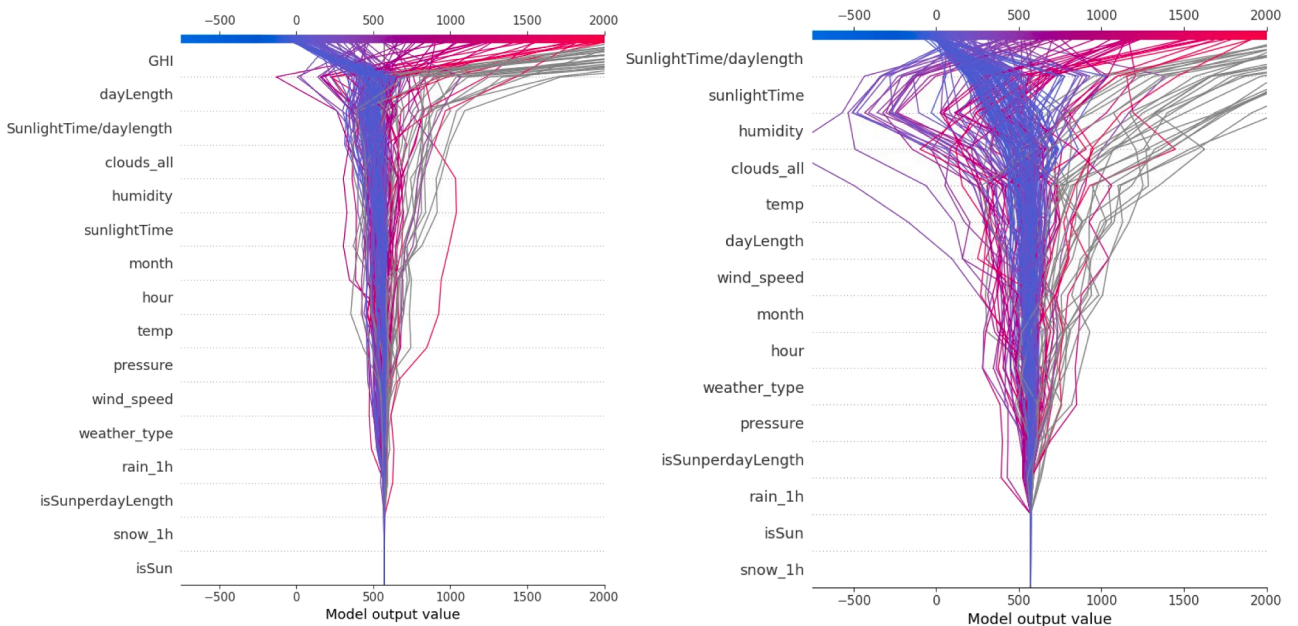


Fig. 5. Evaluation of the impact of 200 data points on the outcome.

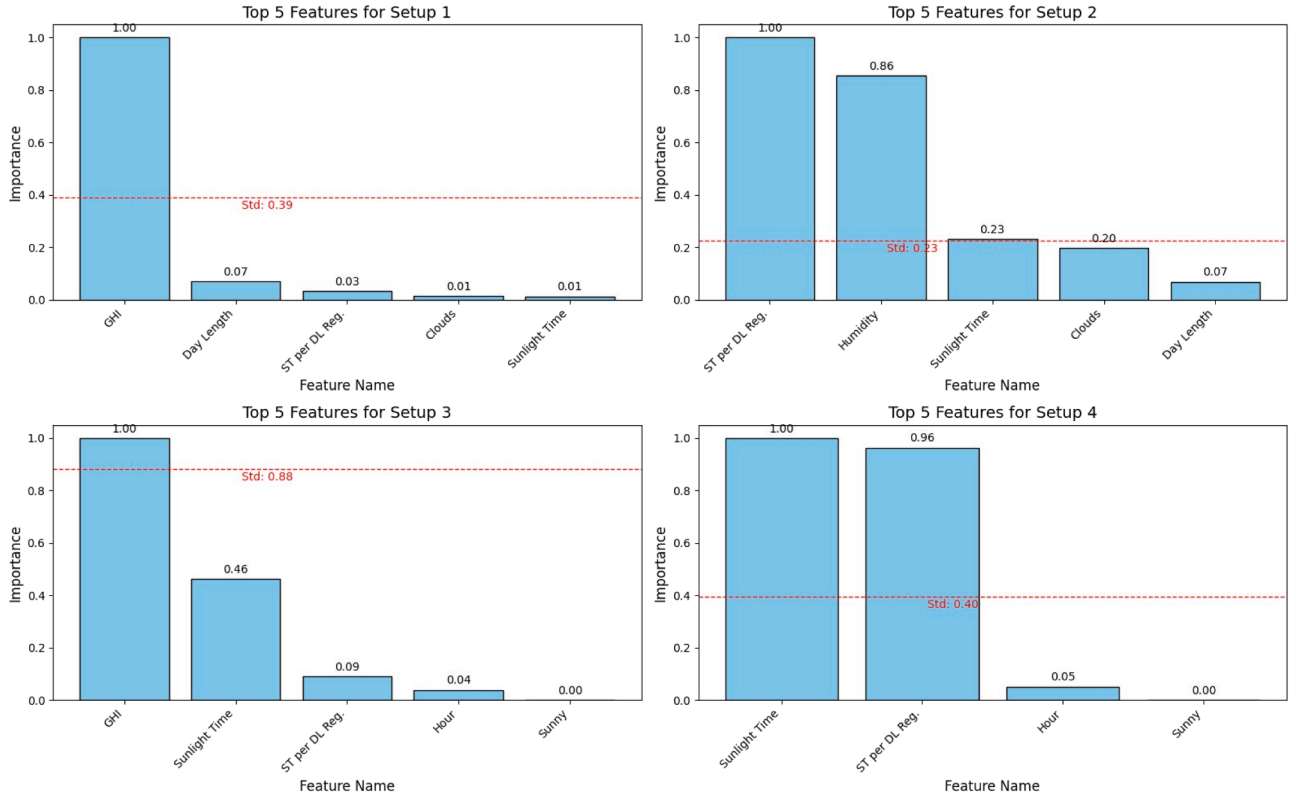


Fig. 6. Normalized results of the Top 5 features for every Setup.

data was randomly altered in each configuration. This procedure was repeated 20 times per dataset. An illustrative example is provided in Table 3, where the introduced noise is reflected in the comparison table.

The results clearly demonstrate that Setup 1 is more sensitive to corruption than Setup 2. In Setup 1, the dominance of the feature GHI cannot be offset by other features, leading to a more pronounced decline in performance. This observation provides empirical evidence that supports the decision to remove dominant features. A single feature that disproportionately influences model forecasts can, when corrupted, trigger instability across the entire model. This effect underscores the importance of mitigating feature dominance to preserve model robustness.

This effect is further substantiated in Fig. 7, which depicts the distribution of validation R^2 values across all 20 corrupted iterations. Setup 1 exhibits greater variance and a higher average deviation from the reference value. In contrast, Setup 2, which features a more balanced configuration, yields an average deviation of 0.008, which is

approximately three times lower than the value observed for Setup 1. This reduction in variance reflects improved robustness and generalization, and supports the hypothesis that eliminating feature dominance enhances resilience under perturbations.

5. Explainable artificial intelligence feature selection framework

Based on our results and findings, we deduced the XAI-FS Framework. The XAI-FS Framework combines performance metrics, cost considerations, and robustness checks into a single, integrated decision-making process. The dataset is defined, and a preliminary statistical analysis is performed to identify redundancies and correlations, ensuring that the original dataset is both comprehensive and manageable. XAI methods such as SHAP and ELI5 are then applied to determine the significance and dominance of features and form subsets for systematic evaluation. Our iterative framework allows practitioners to progressively refine the feature set while maintaining transparency at each decision point. The iterative structure enables features to be excluded and reintroduced in a refined form, allowing for dynamic adaptation to changing data conditions. Each iteration evaluates three critical dimensions: predictive performance, dominance, and cost. These dimensions are integrated into a multi-objective decision function:

$$\max_S J(S) = R^2(S) - \alpha \cdot DI(S) - \gamma \cdot \sum_{i \in S} CostClass(i) \quad (6)$$

where $R^2(S)$ denotes predictive accuracy, $DI(S)$ measures dominance using a Herfindahl–Hirschman-based index, and $CostClass(i)$ represents the acquisition cost of feature i . The parameters α and γ allow decision-makers to balance accuracy, robustness, and economic efficiency according to operational priorities. The objective function is maximized over the subsets S . This formulation ensures that feature selection decisions are not only performance-driven but also cost-aware and auditable. The framework includes documentation and robustness checks to

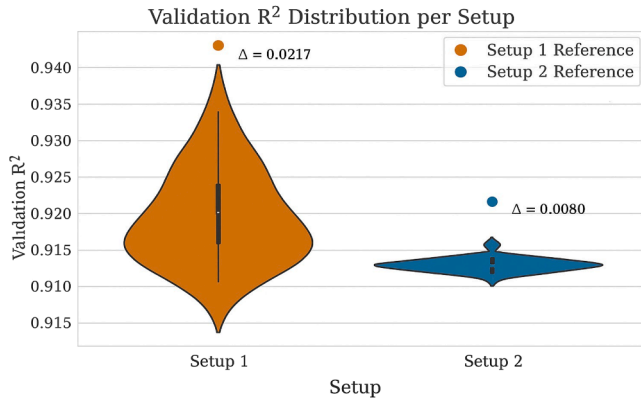


Fig. 7. Distribution of model robustness for each Setup.

enable traceable human-in-the-loop interventions. This design allows domain experts to evaluate or override automated recommendations, ensuring that operational knowledge complements algorithmic insights. Embedded documentation ensures verifiability and traceability of each decision step for third parties. This guarantees a high degree of transparency and audit compliance. The XAI-FS Framework provides a tight interface between automated processes and human expertise, allowing specialist users to evaluate or override the model's recommendations at critical decision points. Furthermore, the modular structure of the framework facilitates integration into MLOps pipelines, making it suitable for industrial deployment where transparency, cost control, and reliability are critical.

Fig. 8 presents the structured framework developed in our study to integrate XAI into feature selection workflows in energy forecasting. Our XAI-FS Framework translates these insights into a structured and repeatable process for feature selection in energy forecasting in practice. Although the XAI-FS Framework was developed in the context of energy forecasting, its modular structure enables its application to other data-intensive areas with high requirements for transparency and cost control.

6. Discussion, recommendations, limitations, and further research

Our framework reveals that the baseline configuration, Setup 1, which utilizes the complete feature set, achieves the highest predictive accuracy, thereby establishing a robust reference point. In Setup 2, the removal of the dominant feature GHI results in only a marginal decline in performance ($R^2 = 0.934$), while substantially redistributing feature importance. This redistribution reduces the dominance index from 0.365 to 0.169, indicating a more balanced contribution across features. Setup 3, excluding cost-intensive low-impact features, shows a slightly greater reduction ($R^2=0.906$) than Setup 2, indicating that removing dominant features is safe, but cumulative removal needs caution. Setup 4, which combines both strategies, leads to underfitting ($R^2 = 0.659$), confirming that excessive feature reduction compromises model reliability. Robustness tests further underscore the benefits of reducing dominance. When noise is introduced to the most dominant feature, Setup 1 experiences a more pronounced decline in R^2 and a greater increase in RMSE compared to Setup 2. This finding highlights that a more balanced feature distribution enhances model stability in the face of data perturbations. Visual evidence from Table 3 and Fig. 7 supports this interpretation, showing narrower variance in robustness outcomes for Setup 2. These results demonstrate that XAI-based feature selection not

only improves interpretability but also enhances robustness by mitigating overreliance on individual predictors.

Recent studies consistently emphasize the importance of feature selection in enhancing interpretability, computational efficiency, and robustness in energy forecasting [19,22,49,88]. Traditional methods like filter, wrapper, and embedded approaches often lack transparency and fail to integrate domain-specific knowledge [19,22,119]. In response to these limitations, related work introduced XAI methods to increase trust in AI-based energy systems [33,38,52,63]. For example, Sim et al. [83] and van Zyl et al. [19] demonstrated that SHAP can improve interpretability in load forecasting, while Maarif et al. [59] applied XAI for parameter analysis in LSTM-based models. However, the systematic integration of XAI-based feature selection into structured frameworks for feature reduction remains insufficiently addressed [19,38]. We contribute to closing this gap by demonstrating that XAI-based feature selection enhances transparency, reduces data-acquisition cost, and maintains predictive accuracy. In addition, the findings advance theoretical discourse on trust and human-in-the-loop approaches in AI [33–35] by demonstrating that XAI facilitates a more balanced distribution of feature importance, thereby increasing robustness under noisy conditions and reducing the dominance of individual features. Unlike Sim et al. [83], who primarily applied XAI to improve interpretability in LSTM-based models, our framework embeds XAI into a structured feature selection framework that explicitly incorporates cost efficiency. Similarly, although van Zyl et al. [19] highlighted the transparency benefits of SHAP, their work did not address systematic feature reduction or robustness under noise, which are central to the contributions of our framework. In contrast to Maarif et al. [59], who focused on parameter-level analysis, we operationalize XAI for feature-level governance, enabling balanced importance distributions and improved model stability. Beyond empirical performance, our framework advances theoretical perspectives on TAI by reducing feature dominance and enhancing robustness, thereby aligning with Durán and Pozzi's [34] criteria for transparency under uncertainty. Documentation and expert oversight reflect human-in-the-loop principles [52], while the cost-aware objective function supports economically efficient modeling [32].

To address RQ1, our XAI-FS Framework employs SHAP and ELI5 to identify dominant and cost-intensive features, enabling targeted reduction of input variables. The experimental setups summarized in Table 2 demonstrate that XAI supports systematic and transparent feature reduction while preserving predictive performance, thereby lowering data acquisition cost and enhancing model interpretability. To address RQ2, we examine the effects of removing the predominant

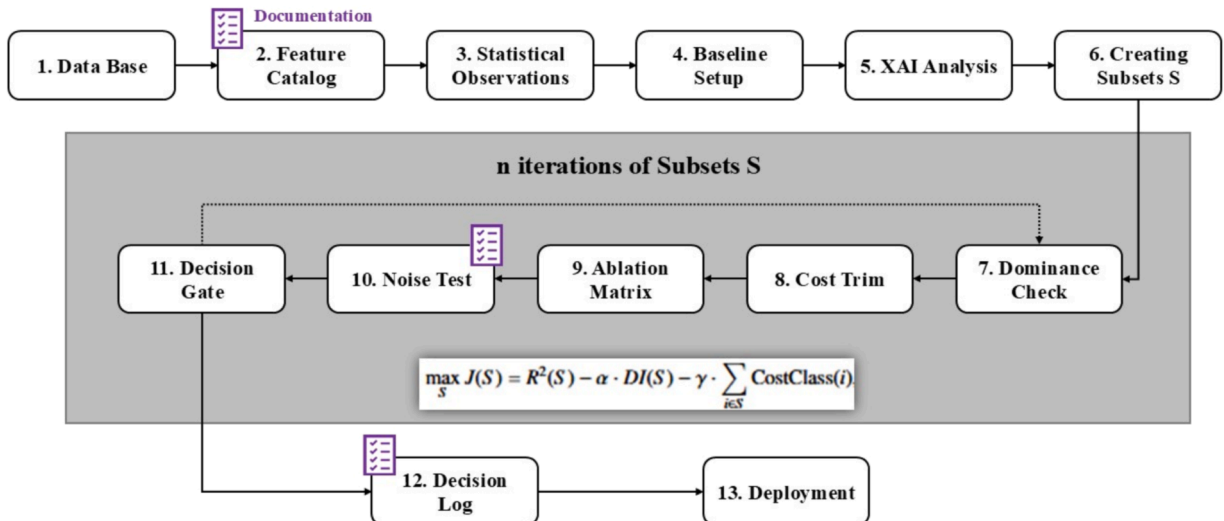


Fig. 8. XAI-FS framework.

features on model behavior and performance. As shown in [Tables 2 and 3](#), excluding the most dominant feature (GHI) in Setup 2 resulted in a substantial decrease in the dominance index, from 0.365 to 0.169. This adjustment improved robustness under noise, as confirmed by the analysis in [Fig. 7](#). Although predictive accuracy declined marginally ($\Delta R^2 \approx 0.009$), the redistribution of feature importance resulted in a more balanced and resilient model structure, mitigating the risk of overreliance on individual features. Finally, RQ3 examines how XAI can facilitate the observation and interpretation of feature importance distributions, thereby enabling a more equitable allocation of relevance across input variables. The results and findings reveal that XAI enhances explainability and operationalizes fairness in feature weighting. [Fig. 8](#) illustrates how the process facilitates governance-ready, cost-aware decisions by making feature relevance transparent and robust. This positions XAI as a key enabler of trustworthy, efficient AI applications in the energy sector.

Several limitations should be acknowledged when interpreting these results and findings. First, our study is based on a single PV dataset, which restricts the generalizability of the results to other energy sources or geographic contexts. Although two additional datasets were used for validation, they were not fully optimized, which may have influenced the comparative performance outcomes. Second, the feature-removal strategy relied on iterative ablation and expert judgment. While our framework ensures transparency, it introduces subjectivity and may not capture all optimal feature subsets. Third, the cost-aware objective function employs a simplified cost classification rather than detailed economic models, which limits its precision for real-world deployment. Finally, the results indicate that aggressive feature reduction (Setup 4) can lead to severe underfitting, emphasizing the need for formal stopping criteria based on performance thresholds and dominance metrics. Further limitations to be considered concern the integration of domain-specific knowledge under uncertainty. Fuzzy logic approaches enable the incorporation of experiential expert knowledge through rule-based systems and linguistic variables. Prior research indicates that fuzzy inference systems can improve robustness and interpretability, particularly in the presence of incomplete or noisy data [[120–123](#)]. In data-scarce domains such as energy forecasting, combining fuzzy logic with physics-informed constraints may further enhance model reliability and transparency.

Further research should investigate the automated integration of XAI into feature selection pipelines to reduce reliance on manual ablation. Validation across diverse energy sources and geographic contexts is crucial for assessing the generalizability of results and findings. Further research addresses real-time robustness of models under concept drift and sensor failures, as well as the role of human factors in interpreting XAI outputs and governance artifacts. Finally, the development of standardized metrics for balancing interpretability, robustness, and cost would support broader adoption of XAI-based feature selection in industrial applications. Further research is needed to examine hybrid models that combine physics-informed constraints with fuzzy-logic-based reasoning. This integration may support the systematic incorporation of expert knowledge under uncertainty and improve robustness, interpretability, and performance in data-limited energy forecasting contexts.

We deduced a transparent, XAI-based framework for feature selection in energy forecasting that integrates interpretability, robustness, and cost-awareness. Unlike related work that focuses only on using XAI methods for interpretability, our framework combines these aspects with systematic feature reduction. It operationalizes the removal of dominant and cost-intensive features through SHAP and ELI5 without compromising performance. This approach goes beyond the conventional use of XAI for explanation by embedding it into a structured decision process for feature governance. Our framework demonstrates that reducing dominance improves robustness under noise, which has not been addressed in prior studies. The inclusion of a cost-sensitive objective function and human-in-the-loop mechanisms supports accountable

and economically efficient model design.

We developed a review in [Table A4](#) that translates the literature into a structured, comparative knowledge matrix. The selected literature was identified based on the added value of the corresponding contributions. The systematic comparison of key literature along consistent parameters (primary contribution, research focus, research gaps, data type, explainability, robustness, cost awareness, and our contributions to each paper) provides a descriptive and consistent presentation of our research. [Table A4](#) allows structural patterns and deficits in the existing literature to be made visible. This enables the identification of research niches, methodological weaknesses, and desiderata in the connection between XAI, feature selection, and energy forecasting. We classify our research individually by the parameters (x-axis) and highlight its contribution to the selected literature (y-axis).

7. Conclusions

The transparency and explainability of AI models are increasingly critical for the successful deployment of AI-based systems. However, feature reduction processes and their selection mechanisms often remain opaque, limiting users' ability to understand the rationale behind feature elimination and to incorporate domain-specific knowledge. We addressed this gap by introducing a transparent, human-in-the-loop feature selection framework grounded in XAI, named the XAI-FS Framework. Our framework enables energy providers to substantially reduce data acquisition and storage cost while maintaining high predictive accuracy, thereby improving operational efficiency and sustainability. At the same time, our framework contributes to academic discourse by bridging the gap between XAI and feature selection, enhancing model robustness and interpretability in the context of energy forecasting. We addressed research objectives by integrating XAI methods, such as SHAP and ELI5, into the feature selection process to enhance transparency and enable users to exert enhanced influence over selection decisions. When combined with expert knowledge, these methods provide insights into the impact of individual features on model performance and illustrate how forecasting changes when features are removed. Our framework leads to a redistribution of feature importance, resulting in a more balanced influence of features. Importantly, overall model quality remains largely unaffected when removing low-impact or cost-intensive features, which supports cost-efficient data strategies. The simultaneous removal of dominant and cost-intensive features can exceed a critical threshold, causing rapid performance deterioration. Our results and findings further demonstrate that reducing feature dominance enhances robustness under noisy conditions, thereby preventing biased data from disproportionately influencing model outcomes. XAI methods enable a comprehensive understanding of feature dependencies and their shifts during feature removal, supporting informed decisions and enhanced user control. This transparency fosters trust, strengthens human-in-the-loop interaction, and ensures that feature selection aligns with both technical and economic objectives. Further research should automate XAI integration into feature selection, validate the framework across diverse energy contexts, and study human factors affecting adoption and trust.

Declaration of generative AI and AI-assisted technologies in the writing process

The authors utilized DeepL for translation and Microsoft Copilot for paraphrasing. After employing these methods, the authors reviewed and edited all content and take full responsibility for the final content.

CRediT authorship contribution statement

Leonard Kost: Writing – original draft, Software, Methodology.
Sarah K. Lier: Writing – original draft, Visualization, Validation.
Michael H. Breitner: Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Appendix

Fig. A9, Table A4

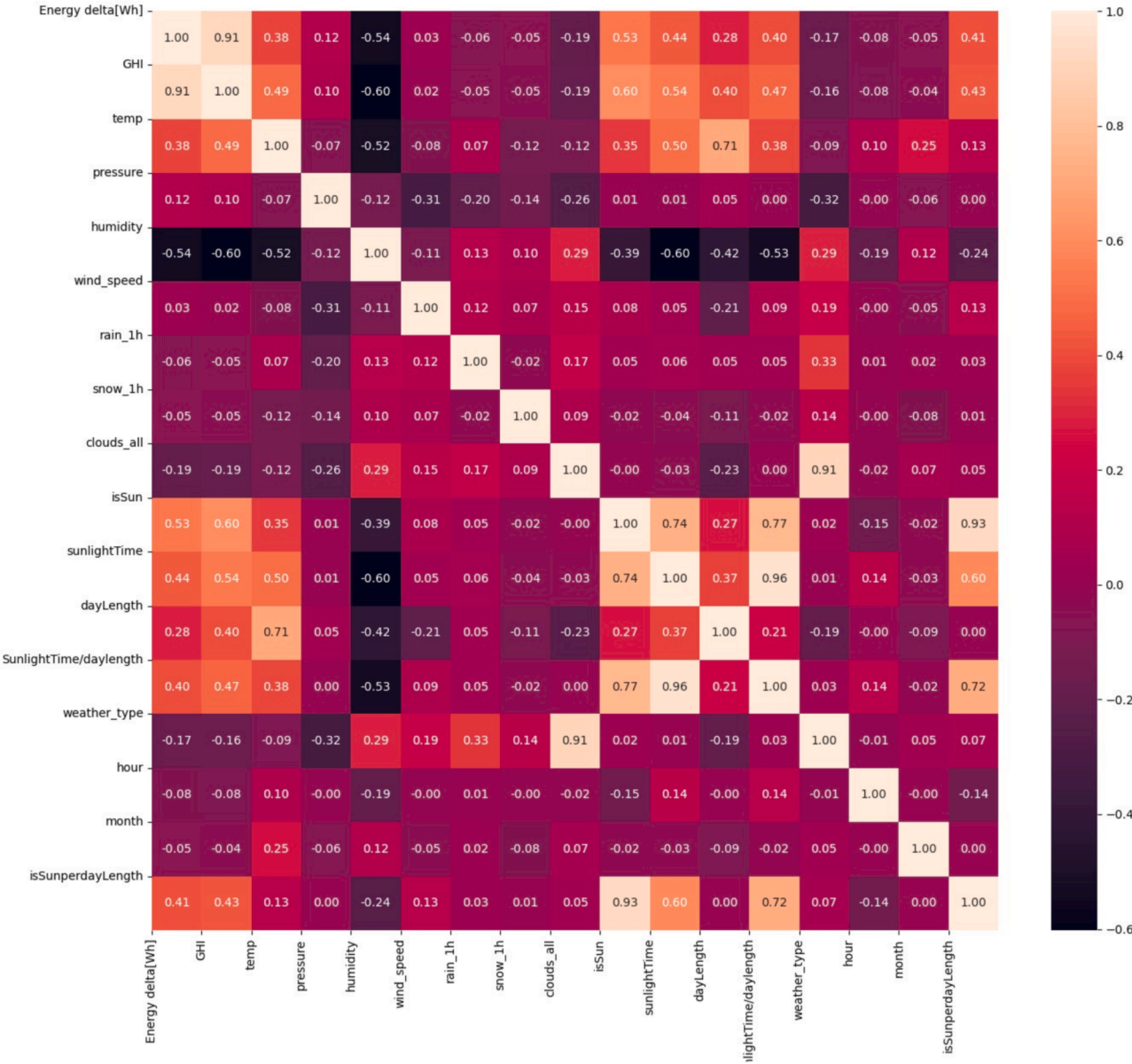


Fig. A9. Correlation of the features.

Table A4

Comparison of literature to our paper.

Authors	Primary Contribution	Research Focus	Identified research needs/ Gaps	Type of data/ Application	Explainability/ XAI	Robustness/ Generalizability	Cost Awareness	Our Contribution to each paper
Maarif et al. [59]	Combination of LSTM and XAI for energy consumption	Accuracy and traceability for industrial energy	Lack of robustness analysis and systematic feature elimination	Time series of industrial energy consumption	Post-hoc (LIME, SHAP)	-	-	Transfers LSTM and XAI into an explainable FS* architecture with (continued on next page)

Table A4 (continued)

Authors	Primary Contribution	Research Focus	Identified research needs/ Gaps	Type of data/ Application	Explainability/ XAI	Robustness/ Generalizability	Cost Awareness	Our Contribution to each paper
Sim et al. [83]	forecasting; focus on interpretability Application of SHAP for variable evaluation in energy forecasts	consumption data Practical explainability of energy consumption models (SHAP)	Lack of economic or dominance analysis of features	Building consumption data	Post-hoc (SHAP)	-	-	robustness validation Based on the SHAP approach, but integrates cost and dominance aspects
van Zyl et al. [19]	Comparison of Grad-CAM vs SHAP (energy forecast)	Comparison of visual and SHAP-based explanations	Lack of stability analysis and cost dimension	Energy consumption time series (CNN, Grad-CAM)	Post-hoc (SHAP/Grad-CAM)	-	-	Unifies XAI, adds robustness and cost dimensions
Yaprakdal and Arisoy [22]	Hybrid FS and LIME/LSTM model	Hybrid XAI for forecasts	Lack of systematic robustness assessment	Electricity load time series (LSTM + LIME)	Post-hoc (LIME, SHAP)	Partially	-	Converts hybrid approaches into a standardizable framework and closes the gap between dominance index and TAI principles
Sweeney et al. [12]	Forecasting trends; uncertainty modeling and integration	Macro trends in forecasting (data-driven, probabilistic, AI)	Lack of methodological integration of explainability in forecasts	Wind and solar forecasts (review)	Mentions need for forecasts	Emphasizes the need for robust forecasts	-	Implements required integration of uncertainty, robustness, and interpretability aspects
Wigger et al. [23]	Industrial AI application in energy systems; empirically validated approach	Practical use of AI in the energy sector	Lack of systematic XAI validation in the energy sector	Real-time industrial processes	Partially interpretable (black box)	Partial empirical validation	Partial economic optimization	Provides a transparent, explainable alternative to industrial black box approaches
Hong et al. [13]	Meta-review of forecasting methods and trends	Research trends in forecasting	Lack of requirements based on integrated pipelines and uncertainty metrics	Wind and solar forecast data (reanalysis, historical)	Theory/ taxonomy	Identifies requirements	-	Implements required integration of forecast and XAI
Blaszczuk et al. [120]	Comparison of fuzzy logic vs cluster models (interpretability)	Physically and heuristically explainable models	Lack of transferability to data-driven models	Thermofluid simulation data; model-based comparative study	Fuzzy logic interpretable	-	-	Integrated interpretable components data-driven
Rusilowati et al. [18]	Overview of AI for energy efficiency	AI and sustainability	Lack of operationalization of transparency	Macro energy data (solar, wind, biomass); conceptual	Overview	-	-	Operationalizes transparency metrics in a measurable way
Detyniecki et al. [41]	PV forecast based on meteorological variables	Classic PV forecast (baseline)	Lack of explainable FS and cost assessment	Meteorological variables (solar radiation, wind, etc.)	-	-	-	Converts weather base models into an explainable FS pipeline
Our Paper	Robust, explainable, cost-conscious FS framework	Holistic integration of XAI, robustness, costs, and FS	Closes gaps through explainable, robust, cost-conscious FS; integrates SHAP/ELI5 into a DSR-based framework	Solar and wind energy forecasts with multiple datasets	SHAP + ELI5 (integrated)	Dominance Index introduced to identify feature sensitivity	Cost integrated into objective function R^2 , dominance index, cost)	

* FS = Feature Selection.

Data availability

Data will be made available on request.

References

- [1] Yolcan OO. World energy outlook and state of renewable energy: 10-year evaluation. *Innov Green Dev* 2023;2. <https://doi.org/10.1016/j.igd.2023.100070>.
- [2] Billewar SR, Londhe G, Mane PS. World energy demand. *Integr Green Energy Solut* 2023;2:275–316.
- [3] Nadeau KC, Agache I, Jutel M, Annesi Maesano I, Akdis M, Sampath V, d'Amato G, Cecchi L, Traidl-Hoffmann C, Akdis CA. Climate change: a call to action for the United Nations. *Allergy* 2022;77:1087–90.
- [4] Matsuo Y, Yanagisawa A, Yamashita Y. A global energy outlook to; 2035 with strategic considerations for Asia and Middle East energy supply and demand interdependencies. *Energy Strategy Rev* 2013;(2):79–91.
- [5] Statistisches Bundesamt (Destatis). Ausbau erneuerbarer energien in Europa. URL: https://www.destatis.de/Europa/DE/Thema/Umwelt-Energie/Ausbau_ErneuerbareEnergien.html; 2024 [accessed 19 September 2025]].
- [6] Dimitriadis CN, Passalis N, Georgiadis MC. A deep learning framework for photovoltaic power forecasting in multiple interconnected countries. *Sustain Energy Technol Assess* 2025;77:#104330. <https://doi.org/10.1016/j.seta.2025.104330>. URL, <https://www.sciencedirect.com/science/article/pii/S2213138825001614>.
- [7] Passalis N, Dimitriadis CN, Georgiadis MC. Residual adaptive input normalization for forecasting renewable energy generation in multiple countries. *Pattern Recognit Lett* 2025;196:52–8. <https://doi.org/10.1016/j.patrec.2025.05.008>. URL, <https://www.sciencedirect.com/science/article/pii/S0167865525001977>.

- [8] Brunner C, Deac G, Braun S, Zöphel C. The future need for flexibility and the impact of fluctuating renewable power generation. *Renew Energy* 2020;149: 1314–24. <https://doi.org/10.1016/j.renene.2019.10.128>.
- [9] Abrell J, Rausch S, Streitberger C. Buffering volatility: storage investments and technology-specific renewable energy support. *Energy Econ* 2019;84. <https://doi.org/10.1016/j.eneco.2019.07.023>.
- [10] de São D, Faria P, Vale Z. Smart energy community: a systematic review with meta-analysis. *Energy Strategy Rev* 2021;36. <https://doi.org/10.1016/j.esr.2021.100678>.
- [11] Ahmad T, Zhang D, Huang C, Zhang H, Dai N, Song Y, Chen H. Artificial intelligence in sustainable energy industry: status quo. *chall oppor* 2021. <https://doi.org/10.1016/j.jclepro.2021.125834>.
- [12] Sweeney C, Bessa RJ, Browell J, Pinson P. The future of forecasting for renewable energy. *Wiley Interdiscip Rev: Energy Environ* 2020;9. <https://doi.org/10.1002/wene.365>.
- [13] Hong T, Pinson P, Wang Y, Weron R, Yang D, Zareipour H. Energy forecasting: a review and outlook. *IEEE open access. J Power Energy* 2020;7:376–88. <https://doi.org/10.1109/OAJPE.2020.3029979>.
- [14] Debnath KB, Mourshed M. Forecasting methods in energy planning models. *Renew Sustain Energy Rev* 2018;88:297–325. <https://doi.org/10.1016/j.rser.2018.02.002>.
- [15] Le TT, Priya JC, Le HC, Le NVL, Duong MT, Cao DN. Harnessing artificial intelligence for data-driven energy predictive analytics: a systematic survey towards enhancing sustainability. *Int J Renew Energy Dev* 2024;13:270–93. <https://doi.org/10.61435/ijred.2024.60119>.
- [16] Szczepaniuk H, Szczepaniuk EK. Applications of artificial intelligence algorithms in the energy sector. *Energies* 2023;16. <https://doi.org/10.3390/en16010347>.
- [17] Y. Liu, O.C. Esan, Z. Pan, L. An, Machine learning for advanced energy materials; 2021. <https://doi.org/10.1016/j.egyai.2021.100049>.
- [18] Rusilowati U, Ngemba HR, Anugrah RW, Fitriani A, Astuti ED. Leveraging AI for superior efficiency in energy use and development of renewable resources such as solar energy, wind, and bioenergy. *Int Trans Artif Intell* 2024;2:114–20. <https://doi.org/10.33050/italic.v2i2.537>.
- [19] van Zyl C, Ye X, Naidoo R. Harnessing explainable artificial intelligence for feature selection in time series energy forecasting: a comparative analysis of Grad-Cam and SHAP. *Appl Energy* 2024;353. <https://doi.org/10.1016/j.apenergy.2023.122079>.
- [20] Sarah MHB, Lier K, Gerlach Jana. Who needs XAI in the energy sector? A framework to upgrade black box explainability. In: *Proceedings of the 44th International Conference on Information Systems*; 2023.
- [21] T. Zhang, G. Strbac. Novel artificial intelligence applications in energy: a systematic review; 2025. <https://doi.org/10.3390/en18143747>.
- [22] Yaprakdal F, Arisoy MV. A multivariate time series analysis of electrical load forecasting based on a hybrid feature selection approach and explainable deep learning. *Appl Sci* 2023;13. <https://doi.org/10.3390/app132312946>.
- [23] Wigger M, Burggräf P, Steinberg F, Becher A, Heinbach B. Integrating artificial intelligence into energy management: a case study on energy consumption data analysis and forecasting in a German manufacturing company. *Energy AI* 2025; 21. <https://doi.org/10.1016/j.egyai.2025.100576>.
- [24] Rai A. Explainable AI: from black box to glass box. *J Acad Mark Sci* 2020;48: 137–41. <https://doi.org/10.1007/s11747-019-00710-5>.
- [25] Hassija V, Chamola V, Mahapatra A, Singal A, Goel D, Huang K, Scardapane S, Spinelli I, Mahmud M, Hussain A. Interpreting black-box models: a review on explainable artificial intelligence. *Cogn Comput* 2024;16:45–74. <https://doi.org/10.1007/s12559-023-10179-8>.
- [26] Zhao R, Xie Z, Zhuang Y, Yu PL. Automated quality evaluation of large-scale benchmark datasets for vision-language tasks. *Int J Neural Syst* 2024;34. <https://doi.org/10.1142/S0129065724500096>.
- [27] Jayawardene V, Sadiq S, Indulska M. An analysis of data quality dimensions. *ITEEE Tech Rep* 2015.
- [28] Picard S, Chapdelaine C, Lefevre B, Jenn E. Ensuring dataset quality for machine learning certification. In: *International Symposium on Software Reliability Engineering Workshops*; 2021. p. 275–82. <https://doi.org/10.1109/ISSREW51248.2020.00085>.
- [29] Slota SC, Fleischmann KR, Greenberg S, Verma N, Cummings B, Li L, Shenefiel C. Good systems, bad data?: interpretations of AI hype and failures. *Proc Assoc Inf Sci Technol* 2020;57. <https://doi.org/10.1002/pra2.275>.
- [30] Haug A, Zachariassen F, van Liempd D. The costs of poor data quality. *J Ind Eng Manag* 2011;4:168–93. <https://doi.org/10.3926/jiem.2011.v4n2.p168-193>.
- [31] Koren M, Peretz O, Koren O. Feature engineering procedure for information enrichment. In: *International Conference on Advanced Enterprise Information System (AEIS)*; 2023. p. 28–34. <https://doi.org/10.1109/AEIS61544.2023.00012>.
- [32] Erion G, Janizek JD, Hudelson C, Utarnachitt RB, Sayre MR, White NJ, Lee S-I. Coai. Cost-aware artificial intelligence for health care. *Nat Biomed Eng* 2021;6. <https://doi.org/10.1101/2021.01.19.21249356>.
- [33] Adadi A, Berrada M. Peeking inside the black-box: a survey on explainable artificial intelligence (XAI). *IEEE Access* 2018;6:52138–60. <https://doi.org/10.1109/ACCESS.2018.2870052>.
- [34] Durán JM, Pozzi G. Trust and trustworthiness in AI. *Philos Technol* 2025;38. <https://doi.org/10.1007/s13347-025-00843-2>.
- [35] Simion M, Kelp C. Trustworthy artificial intelligence. *Asian J Philos* 2023;2. <https://doi.org/10.1007/s44204-023-00063-5>.
- [36] Yang G, Ye Q, Xia J. Unbox the black-box for the medical explainable AI via multi-modal and multi-centre data fusion: a mini-review, two showcases and beyond. *Inf Fusion* 2022;77:29–52. <https://doi.org/10.1016/j.inffus.2021.07.016>.
- [37] D. Kumar, G.W. Taylor, A. Wong. Opening the black box of financial AI with clear-trade: a class-enhanced attentive response approach for explaining and visualizing deep learning-driven stock market prediction; 2017. [10.48550/arXiv.1709.01574](https://arxiv.org/abs/1709.01574).
- [38] Machlev R, Heistrene L, Perl M, Levy K, Belikov J, Mannor S, Levron Y. Explainable artificial intelligence (XAI) techniques for energy and power systems: review, challenges and opportunities. *Energy AI* 2022;9. <https://doi.org/10.1016/j.egyai.2022.100169>. URL, <https://www.sciencedirect.com/science/article/pii/S2666546822000246>.
- [39] Heistrene L, Machlev R, Perl M, Belikov J, Baimel D, Levy K, Mannor S, Levron Y. Explainability-based trust algorithm for electricity price forecasting models. *Energy AI* 2023;14. <https://doi.org/10.1016/j.egyai.2023.100259>.
- [40] Lier SK, Eppers T, Gerlach J, Müller P, Breiter MH. An iterative five-phases process model to successfully implement AI for cybersecurity in a corporate environment. *Electron Mark* 2025. <https://doi.org/10.1007/s12525-025-00802-x>.
- [41] Detyniecki M, Marsala C, Krishnan A, Siegel M. Weather-based solar energy prediction. In: *IEEE International Conference on Fuzzy Systems*; 2012. <https://doi.org/10.1109/FUZZ-IEEE.2012.6251145>.
- [42] Sharadga H, Hajimirza S, Balog RS. Time series forecasting of solar power generation for large-scale photovoltaic plants. *Renew Energy* 2020;150:797–807. <https://doi.org/10.1016/j.renene.2019.12.131>.
- [43] Wang RY, Strong DM. Beyond accuracy: what data quality means to data consumers. *J Manag Inf Syst* 1996;12:5–33.
- [44] Goodhue DL, Wybo MD, Kirsch LJ. The impact of data integration on the costs and benefits of information systems. *MIS Q* 1992;16. <https://doi.org/10.2307/249530>.
- [45] Samsi S, Zhao D, McDonald J, Li B, Michaleas A, Jones M, Bergeron W, Kepner J, Tiwari D, Gadepally V. From words to watts: benchmarking the energy costs of large language model inference. In: *High Performance Extreme Computing Conference*; 2023. <https://doi.org/10.1109/HPEC58863.2023.10363447>.
- [46] B. Cottier, R. Rahman, L. Fattorini, N. Maslej, D. Owen. The rising costs of training frontier AI models. 2024. <https://doi.org/10.48550/arXiv.2405.21015>.
- [47] Liu H, Motoda H. Computational methods of feature selection. Chapman & Hall/CRC; 2008.
- [48] Vergara JR, Estévez PA. A review of feature selection methods based on mutual information. *Neural Comput Appl* 2015;24. <https://doi.org/10.1007/s00521-013-1368-0>.
- [49] Li J, Cheng K, Wang S, Morstatter F, Trevino RP, Tang J, Liu H. Feature selection: a data perspective. *ACM Comput Surv* 2017;50. <https://doi.org/10.1145/3136625>.
- [50] Guyon I, Elisseeff A. An introduction to variable and feature selection. *J Mach Learn Res* 2003;3:1157–82.
- [51] Covert IC, Lundberg S, Lee S-I. Understanding global feature contributions with additive importance measures. *Adv Neural Inf Process Syst* 2020;33:17212–23.
- [52] Arrieta AB, Díaz-Rodríguez N, Ser JD, Bannetot A, Tabik S, Barbado A, García S, Gil-Lopez S, Molina D, Benjamins R, Chatila R, Herrera F. Explainable artificial intelligence XAI: concepts, taxonomies, opportunities and challenges toward responsible AI. *Inf Fusion* 2020;58.
- [53] Haug A. Understanding the differences across data quality classifications: a literature review and guidelines for future research. *Int Manag Data Syst* 2021; 121:2651–71. <https://doi.org/10.1108/IMDS-12-2020-0756>.
- [54] Lundberg SM, Lee S-I. A unified approach to interpreting model predictions. *Adv Neural Inf Process Syst* 2017;30.
- [55] Ribeiro MT, Singh S, Guestrin C. Why should I trust you? explaining the predictions of any classifier. In: *Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. 13–17. Association for Computing Machinery; 2016. p. 1135–44. <https://doi.org/10.1145/2939672.2939778>.
- [56] Korobov M, Lopuhin K. Eli5: a python library for debugging/inspecting machine learning classifiers and explaining their predictions. URL, <https://eli5.readthedocs.io/en/latest/>; 2017 [accessed 19 September 2025].
- [57] Lundberg SM, Erion G, Chen H, DeGrave A, Prutkin JM, Nair B, Katz R, Himmelfarb J, Bansal N, Lee SI. From local explanations to global understanding with explainable AI for trees. *Nat Mach Intell* 2020;2:56–67. <https://doi.org/10.1038/s42256-019-0138-9>.
- [58] Chakraborty D, Alam A, Chaudhuri S, Başağaoğlu H, Sulbaran T, Langar S. Scenario-based prediction of climate change impacts on building cooling energy consumption with explainable artificial intelligence. *Appl Energy* 2021;291. <https://doi.org/10.1016/j.apenergy.2021.116807>.
- [59] Maarif MR, Saleh AR, Habibi M, Fitriyani NL, Syafrudin M. Energy usage forecasting model based on long short-term memory (LSTM) and explainable artificial intelligence (XAI). *Information* 2023;14. <https://doi.org/10.3390/info14050265>.
- [60] Mukelabai MD, Wijayantha KG, Blanchard RE. Using machine learning to expound energy poverty in the global south: understanding and predicting access to cooking with clean energy. *Energy AI* 2023;14. <https://doi.org/10.1016/j.egyai.2023.100290>.
- [61] Molnar C. Interpretable machine learning: a guide for making black box models explainable. URL, <https://christophm.github.io/>; 2018 [accessed 19 September 2025].
- [62] Kawakura S, Hirafuji M, Ninomiya S, Shibasaki R. Adaptations of explainable artificial intelligence (XAI) to agricultural data models with eli5, pdpbox, and

- skater using diverse agricultural worker data. *Eur J Artif Intell Mach Learn* 2022; 1:27–34. <https://doi.org/10.24018/ejai.2022.1.3.14>.
- [63] Kuzlu M, Cali U, Sharma V, Güler Ö. Gaining insight into solar photovoltaic power generation forecasting utilizing explainable artificial intelligence tools. *IEEE Access* 2020;8:187814–23. <https://doi.org/10.1109/ACCESS.2020.3031477>.
- [64] Janček AGK, Gansterer WN, Demel MA, Ecker GF. On the relationship between feature selection and classification accuracy. *New Chall Feature Sel Data Min Knowl Discov* 2008;4:90–105.
- [65] Hevner AR, March ST, Park J, Ram S. Design science in information systems research. *MIS Q* 2004;75–105.
- [66] Schoormann T, Möller F, Chandra Kruse L, Otto B. BAUSTEIN—A design tool for configuring and representing design research. *Inf Syst J* 2024;34:1871–901.
- [67] Iheanetu KJ. Solar photovoltaic power forecasting: a review. *Sustainability* 2022; 14. <https://doi.org/10.3390/su142417005>.
- [68] vom Brocke J, Simons A, Riemer K, Niehaves B, Plattfaut R. Standing on the shoulders of giants: challenges and recommendations of literature search in information systems research. *Commun Assoc Inf Syst* 2015;37. <https://doi.org/10.17705/1CAIS.03709>.
- [69] Webster J, Watson RT. Analyzing the past to prepare for the future: writing a literature review. *MIS Q* 2002;26:xiii–xxiii.
- [70] Samarajeewa C, Silva DD, Manic M, Mills N, Moraliyage H, Alahakoon D, Jennings A. An artificial intelligence framework for explainable drift detection in energy forecasting. *Energy AI* 2024;17. <https://doi.org/10.1016/j.egyai.2024.100403>.
- [71] C. Xu, Z. Liao, C. Li, X. Zhou, R. Xie. Review on interpretable machine learning in smart grid. 2022. [doi:10.3390/en15124427](https://doi.org/10.3390/en15124427).
- [72] Jacovi A, Marasović A, Miller T, Goldberg Y. Formalizing trust in artificial intelligence: prerequisites, causes and goals of human trust in AI. Association for Computing Machinery, Inc.; 2021. p. 624–35. <https://doi.org/10.1145/3442188.3445923>.
- [73] O. Arreche, T. Guntur, M. Abdallah. XAI-based feature selection for improved network intrusion detection systems. 2024. [doi:https://doi.org/10.48550/arXiv.2410.10050](https://doi.org/10.48550/arXiv.2410.10050).
- [74] T.Z. Phyu, N.N. Oo. Performance comparison of feature selection methods. 42; 2024. 10.1051/C.
- [75] Michalakopoulos V, Sarmas E, Papias I, Skaloumpakas P, Marinakis V, Doukas H. A machine learning-based framework for clustering residential electricity load profiles to enhance demand response programs. *Appl Energy* 2024;361. <https://doi.org/10.1016/j.apenergy.2024.122943>.
- [76] C.D. Wirz, J.L. Demuth, A. Bostrom, M.G. Cains, I. Ebert-Uphoff, D.J. Gagne, A. Schumacher, A. McGovern, D. Madlambayan. (Re)conceptualizing trustworthy AI: a foundation for change. 2025. [doi:10.1016/j.artint.2025.104309](https://doi.org/10.1016/j.artint.2025.104309).
- [77] Jobin A, Ienca M, Vayena E. The global landscape of AI ethics guidelines. *Nat Mach Intell* 2019;1:389–99. <https://doi.org/10.1038/s42256-019-0088-2>.
- [78] D. Kaur, S. Uslu, K.J. Rittichier, A. Durreli. Trustworthy artificial intelligence: a review; 2023. [doi:10.1145/3491209](https://doi.org/10.1145/3491209).
- [79] Sarah MHB, Lier K, Gerlach Jana. What is ethical AI?—design guidelines and principles in the light of different regions, countries, and cultures. In: Proceedings of the 57th Hawaii International Conference on System Sciences; 2024. p. 6848–57.
- [80] Liu Y, Yao Y, Ton J-F, Zhang X, Guo R, Cheng H, Klovchov Y, Taufiq MF, Li H. Trustworthy LLMs: a survey and guideline for evaluating large language models' alignment. URL, <http://arxiv.org/abs/2308.05374>; 2024.
- [81] Trebbien J, Gorjão LR, Praktikno A, Schäfer B, Witthaut D. Understanding electricity prices beyond the merit order principle using explainable AI. *Energy AI* 2023;13. <https://doi.org/10.1016/j.egyai.2023.100250>.
- [82] Roscher R, Bohn B, Duarte MF, Garcke J. Explainable machine learning for scientific insights and discoveries. *IEEE Access* 2020;8:42200–16. <https://doi.org/10.1109/ACCESS.2020.2976199>.
- [83] Sim T, Choi S, Kim Y, Youn SH, Jang DJ, Lee S, Chun CJ. Explainable AI (XAI)-based input variable selection methodology for forecasting energy consumption. *Electron (Basel)* 2022;11. <https://doi.org/10.3390/electronics11182947>.
- [84] Nasr-Azadani MM, Chatelain J-L. The journey to trustworthy AI: pursuit of pragmatic frameworks. URL, <http://arxiv.org/abs/2403.15457>; 2025.
- [85] B. Li, P. Qi, B. Liu, S. Di, J. Liu, J. Pei, J. Yi, B. Zhou. Trustworthy AI: from principles to practices. 2023. [doi:10.1145/3555803](https://doi.org/10.1145/3555803).
- [86] P. Mikalef, K. Conboy, J.E. Lundström, A. Popović. Thinking responsibly about responsible AI and 'the dark side' of AI. 2022. [doi:10.1080/0960085X.2022.2026621](https://doi.org/10.1080/0960085X.2022.2026621).
- [87] Scharowski N, Benk M, Kühne SJ, Wettstein L, Brühlmann F. Certification labels for trustworthy AI: insights from an empirical mixed-method study. *Assoc Comput Mach* 2023;248–60. <https://doi.org/10.1145/3593013.3593994>.
- [88] Wang H, Liang Q, Hancock JT, Khoshgoftaar TM. Feature selection strategies: a comparative analysis of SHAP-value and importance-based methods. *J Big Data* 2024;11. <https://doi.org/10.1186/s40537-024-00905-w>.
- [89] Marcellio WE, Eler DM. From explanations to feature selection: assessing shap values as feature selection mechanism. Institute of Electrical and Electronics Engineers Inc.; 2020. p. 340–7. <https://doi.org/10.1109/SIBGRAP151738.2020.00053>.
- [90] Teneggi J, Luster A, Sulam J. Fast hierarchical games for image explanations. *IEEE Trans Pattern Anal Mach Intell* 2023;45:4494–503. <https://doi.org/10.1109/TPAMI.2022.3189849>.
- [91] Ribeiro MT, Singh S, Guestrin C. Anchors: high-precision model-agnostic explanations. In: Proceedings of the AAAI Conference on Artificial Intelligence. 31; 2018. <https://doi.org/10.1609/aaai.v32i1.11491>.
- [92] G. Vilone, L. Longo. Explainable artificial intelligence: a systematic review. 2020. [10.48550/arXiv.2006.00093](https://doi.org/10.48550/arXiv.2006.00093).
- [93] Fryer D, Strümke I, Nguyen H. Shapley values for feature selection: the good, the bad, and the axioms. *IEEE Access* 2021;9:144352–60. <https://doi.org/10.1109/ACCESS.2021.3119110>.
- [94] S.M. Lundberg, G.G. Erion, S.-I. Lee. Consistent individualized feature attribution for tree ensembles. 2018. [10.48550/arXiv.1802.03888](https://doi.org/10.48550/arXiv.1802.03888).
- [95] Conte F, D'Antoni F, Natrella G, Merone M. A new hybrid AI optimal management method for renewable energy communities. *Energy AI* 2022;10. <https://doi.org/10.1016/j.egyai.2022.100197>.
- [96] Nabila UM, Lin L, Zhao X, Gurecky WL, Ramuhalli P, Radaideh MI. Data efficiency assessment of generative adversarial networks in energy applications. *Energy AI* 2025;20. <https://doi.org/10.1016/j.egyai.2025.100501>.
- [97] Panjapornpon C, Bardeeniz S, Hussain MA, Chomchai P. Explainable deep transfer learning for energy efficiency prediction based on uncertainty detection and identification. *Energy AI* 2023;12. <https://doi.org/10.1016/j.egyai.2022.100224>.
- [98] Cameron AC, Windmeijer FA. An r-squared measure of goodness of fit for some common nonlinear regression models. *J Econ* 1997;77:329–42. [https://doi.org/10.1016/S0304-4076\(96\)01818-0](https://doi.org/10.1016/S0304-4076(96)01818-0).
- [99] Chai T, Draxler RR. Root mean square error (RMSE) or mean absolute error (MAE)? *Geosci Model Dev* 2014;7. <https://doi.org/10.5194/gmd-7-1525-2014>.
- [100] Hodson TO. Root-mean-square error (RMSE) or mean absolute error (MAE): when to use them or not. *Geosci Model Dev* 2022;15:5481–7. <https://doi.org/10.5194/gmd-15-5481-2022>.
- [101] Enami S. Renewable energy and weather conditions. URL, <https://www.kaggle.com/datasets/samanenami/renewable-energy-and-weather-conditions>; 2024 [accessed 19 September 2025].
- [102] Peng H, Long F, Ding C. Feature selection based on mutual information criteria of max-dependency, max-relevance, and min-redundancy. *IEEE Trans Pattern Anal Mach Intell* 2005;27:1226–38.
- [103] Sekeroglu B, Ever YK, Dimililer K, Al-Turjman F. Comparative evaluation and comprehensive analysis of machine learning models for regression problems. *Data Intell* 2022;4:620–52. https://doi.org/10.1162/dint_a_00155. URLarXiv, https://direct.mit.edu/dint/article-pdf/4/3/620/2039767/dint_a_00155.pdf.
- [104] S. Gooljar, K. Manohar, P. Hosein. Performance evaluation and comparison of a new regression algorithm, arXiv preprint arXiv:2306.09105; 2023.
- [105] Tibshirani R. Regression shrinkage and selection via the lasso. *J R Stat Soc* 1996; 58:267–88.
- [106] A.E. Hoerl, R.W. Kennard. American society for quality ridge regression: biased estimation for nonorthogonal problem. 1970.
- [107] Fix E. Discriminatory analysis: nonparametric discrimination, consistency properties. *USAF Sch Aviat Med* 1985;1.
- [108] Ke G, Meng Q, Finley T, Wang T, Chen W, Ma W, Ye Q, Liu T-Y. LightGBM: a highly efficient gradient boosting decision tree. *Adv Neural Inf Process Syst* 2017; 30.
- [109] Breiman L. Random forests. *Mach Learn* 2001;45:5–32.
- [110] Geurts P, Ernst D, Wehenkel L. Extremely randomized trees. *Mach Learn* 2006;63: 3–42. <https://doi.org/10.1007/s10994-006-6226-1>.
- [111] Friedman JH. Greedy function approximation: a gradient boosting machine. *Ann Stat* 2001;29:1189–232.
- [112] Chen T, Guestrin C. XGBoost: a scalable tree boosting system. In: Proceedings of the ACM SIGKDD international conference on knowledge discovery and data mining. 13–17. Association for Computing Machinery; 2016. p. 785–94. <https://doi.org/10.1145/2939672.2939785>.
- [113] Rumelhart DE, Hinton GE, Williams RJ. Learning internal representations by error propagation, parallel distributed processing, explorations in the microstructure of cognition, ed. de rumelhart and j. mccllelland. 1; 1986. *Biometrika* 1986;71:#6.
- [114] Dietterich T. Overfitting and undercomputing in machine learning. *ACM Comput Surv* 1995;27:326–7.
- [115] Bergstra J, Bengio Y. Random search for hyper-parameter optimization. *J Mach Learn Res* 2012;13:281–305.
- [116] González-Arjona D, Domínguez MM, López-Pérez G, Mulder WH. Noise-injection as an approach to generating random data sets for online tests and virtual labs. *J Lab Chem Educ* 2021;26–35. <https://doi.org/10.5923/j.jlce.20210902.02>.
- [117] Stanczyk U. Feature evaluation by filter, wrapper, and embedded approaches. Berlin, Heidelberg: Springer; 2014. https://doi.org/10.1007/978-3-662-45620-0_3. URL, https://link.springer.com/chapter/10.1007/978-3-662-45620-0_3.
- [118] Das S. Filters, wrappers and a boosting-based hybrid for feature selection. In: Proceedings of the Eighteenth International Conference on Machine Learning (ICML); 2001. p. 74–81. URL, <https://people.csail.mit.edu/sanmay/papers/icml01.pdf>.
- [119] Kaur A, Guleria K, Trivedi NK. Feature selection in machine learning: methods and comparison. In: 2021 International conference on advance computing and innovative technologies in engineering, Institute of Electrical and Electronics Engineers inc; 2021. p. 789–95. <https://doi.org/10.1109/ICACITE51222.2021.9404623>.
- [120] Błaszczyk A, Krzywański J. A comparison of fuzzy logic and cluster renewal approaches for heat transfer modeling in a 1296 t/h cfb boiler with low level of flue gas recirculation. *Arch Thermodyn* 2017;38:91–122.

- [121] Sosnowski M, Krzywanski J, Scurek R. A fuzzy logic approach for the reduction of mesh-induced error in CFD analysis: a case study of an impinging jet. *Entropy* 2019;21:# 1047.
- [122] Chimatapu R, Hagrass H, Starkey A, Owusu G. Explainable AI and fuzzy logic systems. In: *International Conference on Theory and Practice of Natural Computing*. Springer; 2018. p. 3–20.
- [123] Ferreyra E, Hagrass H, Kern M, Owusu G. Depicting decision-making: a type-2 fuzzy logic based explainable artificial intelligence system for goal-driven simulation in the workforce allocation domain. In: *2019 IEEE International Conference on Fuzzy Systems (FUZZ-IEEE)*. IEEE; 2019. p. 1–6.